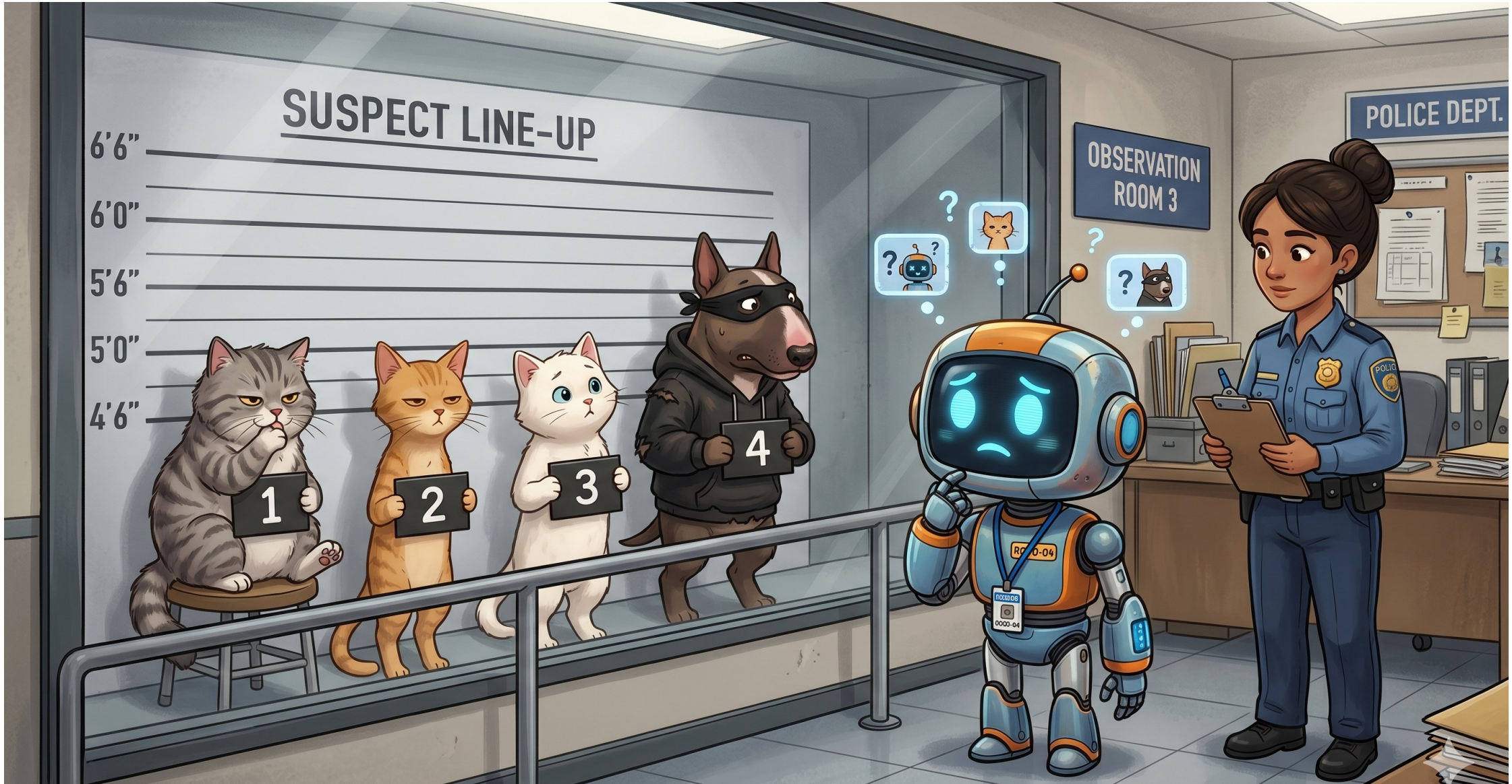


# CS-229

Self supervised vision  
nets intro

Confidence in neural



# Inductive biases in vision: invariance and equivariance

# Invariance

- A function  $f[x]$  is **invariant** to a transformation  $t[]$  if:

$$f[t[x]] = f[x]$$

i.e., the function output is the same even after the transformation is applied.



# Invariance example

e.g., Image classification

- Image has been translated, but we want our classifier to give the same result



# Equivariance

- A function  $f[x]$  is **equivariant** to a transformation  $t[]$  if:

$$f[t[x]] = t[f[x]]$$

i.e., the output is transformed in the same way as the input

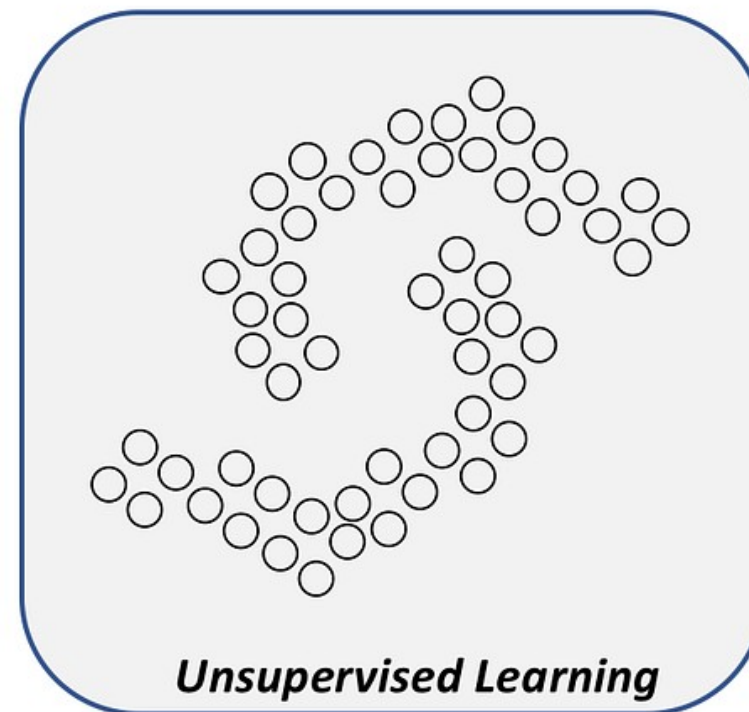
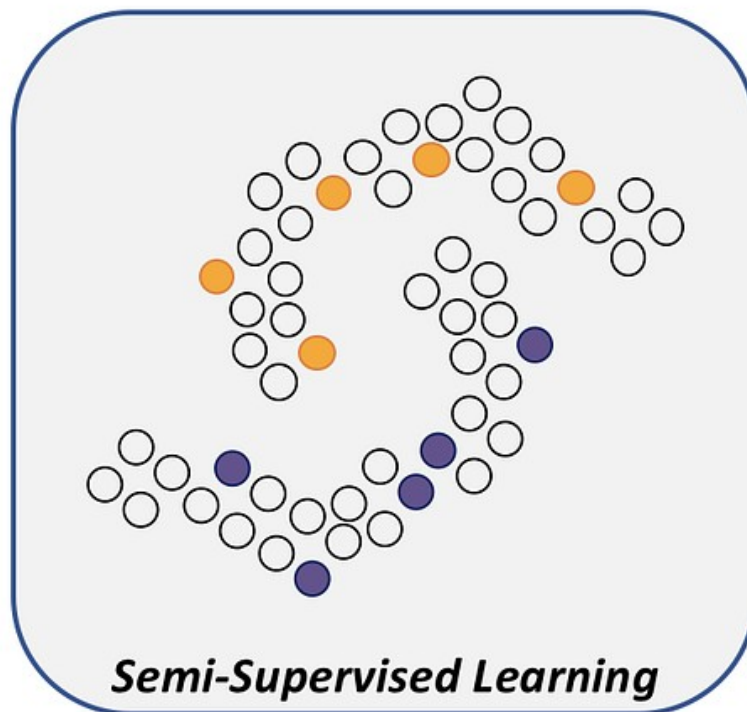
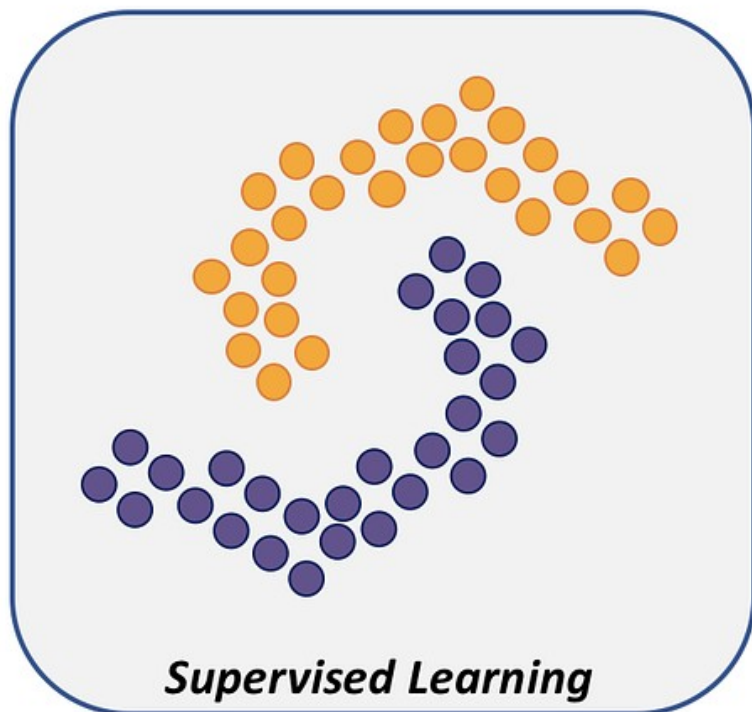
# Equivariance example

e.g., Image segmentation

- Image has been translated and we want segmentation to translate with it



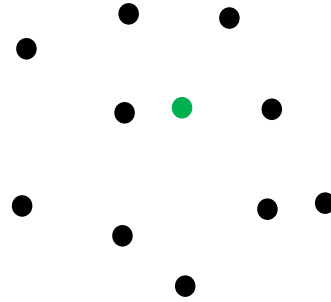
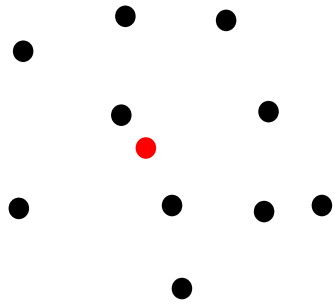
# Semi-supervised Vision



How to exploit the large number of unlabeled examples given in semi-supervised learning?



# Semi-supervised learning with pseudo labeling and augmenting



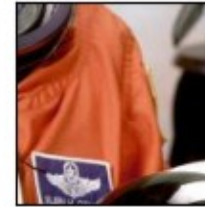
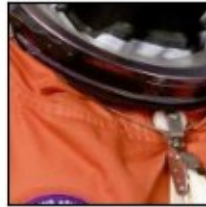
# Data augmentation in computer vision

<https://pytorch.org/vision/stable/transforms.html>

Very useful in self/semi-supervision, but a few problems:

- Hard to replicate this success in non-vision domains
- We should be a little careful, because some augmentation could change the label.

Original image



Original image



Original image



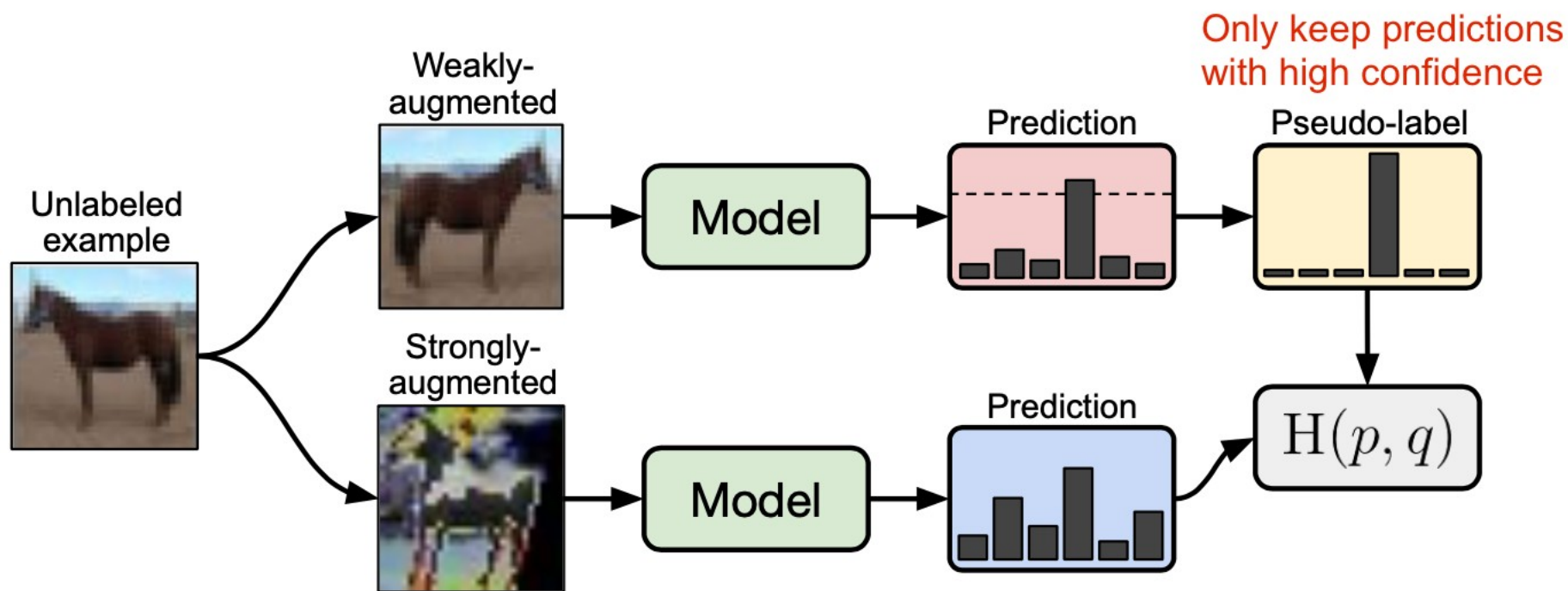
# FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence

Kihyuk Sohn\* David Berthelot\* Chun-Liang Li Zizhao Zhang Nicholas Carlini

Ekin D. Cubuk Alex Kurakin Han Zhang Colin Raffel

Google Research

{kihyuks, dberth, chunliang, zizhaoz, ncarlini,  
cubuk, kurakin, zhanghan, craffel}@google.com



# MixMatch: A Holistic Approach to Semi-Supervised Learning

David Berthelot  
Google Research  
dberth@google.com

Nicholas Carlini  
Google Research  
ncarlini@google.com

Ian Goodfellow  
Work done at Google  
ian-academic@mailfence.com

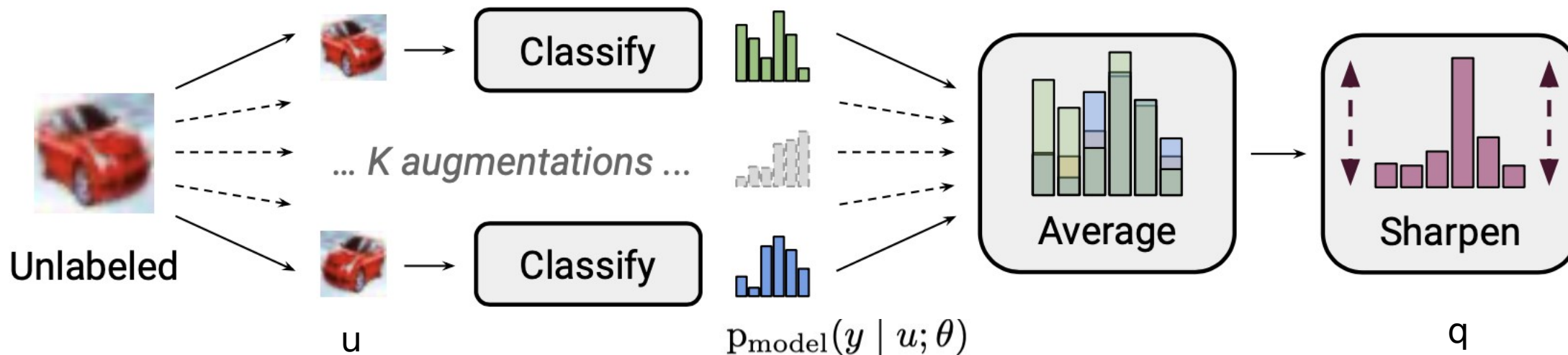
Avital Oliver  
Google Research

Nicolas Papernot  
Google Research

Colin Raffel  
Google Research

Loss = cross entropy (on labeled examples) +

$$\mathcal{L}_u = \frac{1}{L|\mathcal{U}'|} \sum_{u, q \in \mathcal{U}'} \|q - p_{\text{model}}(y | u; \theta)\|_2^2$$



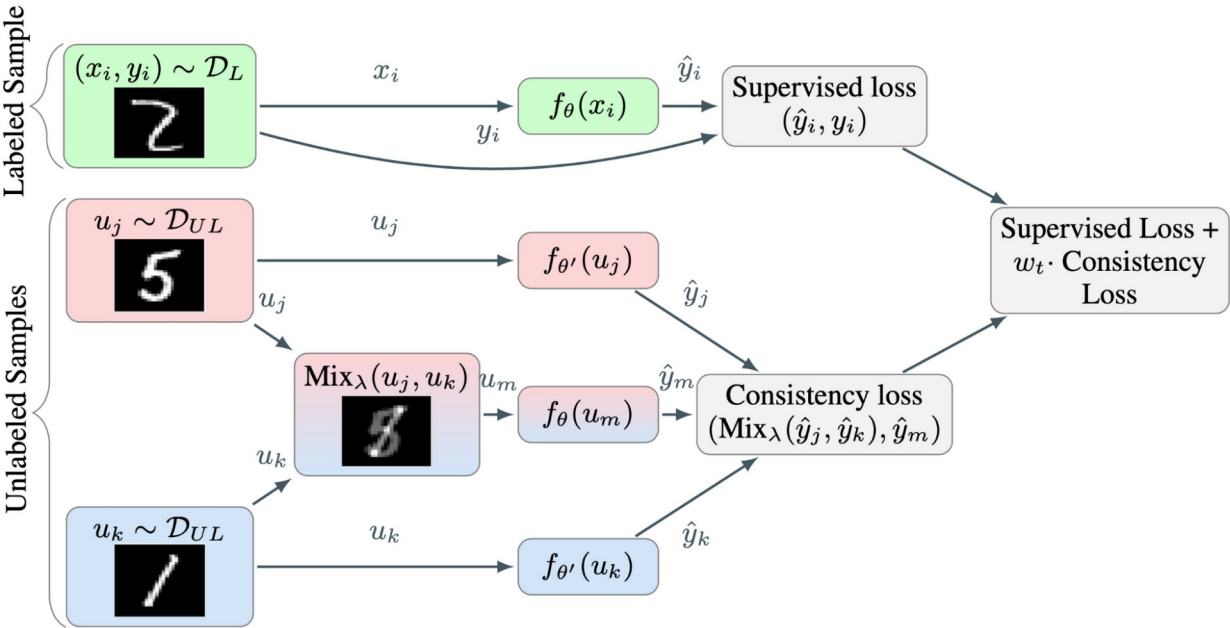
1. *Consistency regularization*: Encourage the model to output the same predictions on perturbed unlabeled samples.
2. *Entropy minimization*: Encourage the model to output confident predictions on unlabeled data.
3. *MixUp* augmentation: (coming up)



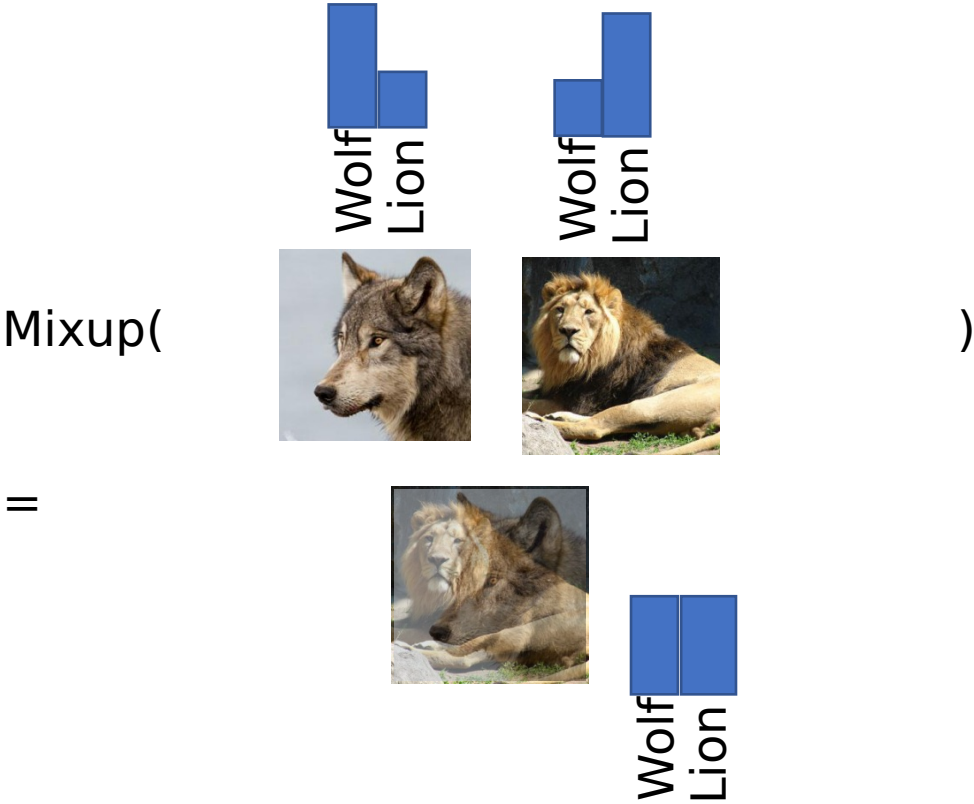
# *mixup*: BEYOND EMPIRICAL RISK MINIMIZATION

Hongyi Zhang  
MIT

Moustapha Cisse, Yann N. Dauphin, David Lopez-Paz\*  
FAIR



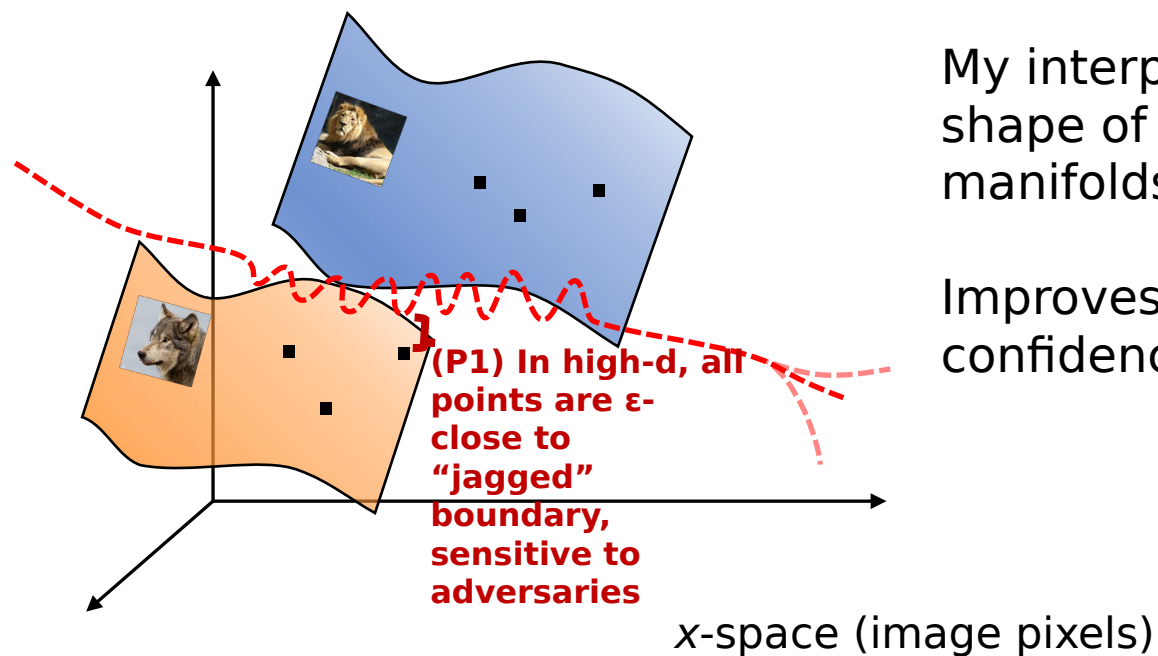
$$\text{mixup}_\lambda(\mathbf{x}_i, \mathbf{x}_j) = \lambda \mathbf{x}_i + (1 - \lambda) \mathbf{x}_j$$
$$p(\text{mixup}_\lambda(y \mid \mathbf{x}_i, \mathbf{x}_j)) \approx \lambda p(y \mid \mathbf{x}_i) + (1 - \lambda) p(y \mid \mathbf{x}_j)$$



# *mixup*: BEYOND EMPIRICAL RISK MINIMIZATION

Hongyi Zhang  
MIT

Moustapha Cisse, Yann N. Dauphin, David Lopez-Paz\*  
FAIR



My interpretation – mixup constrains the shape of the classifier between these manifolds

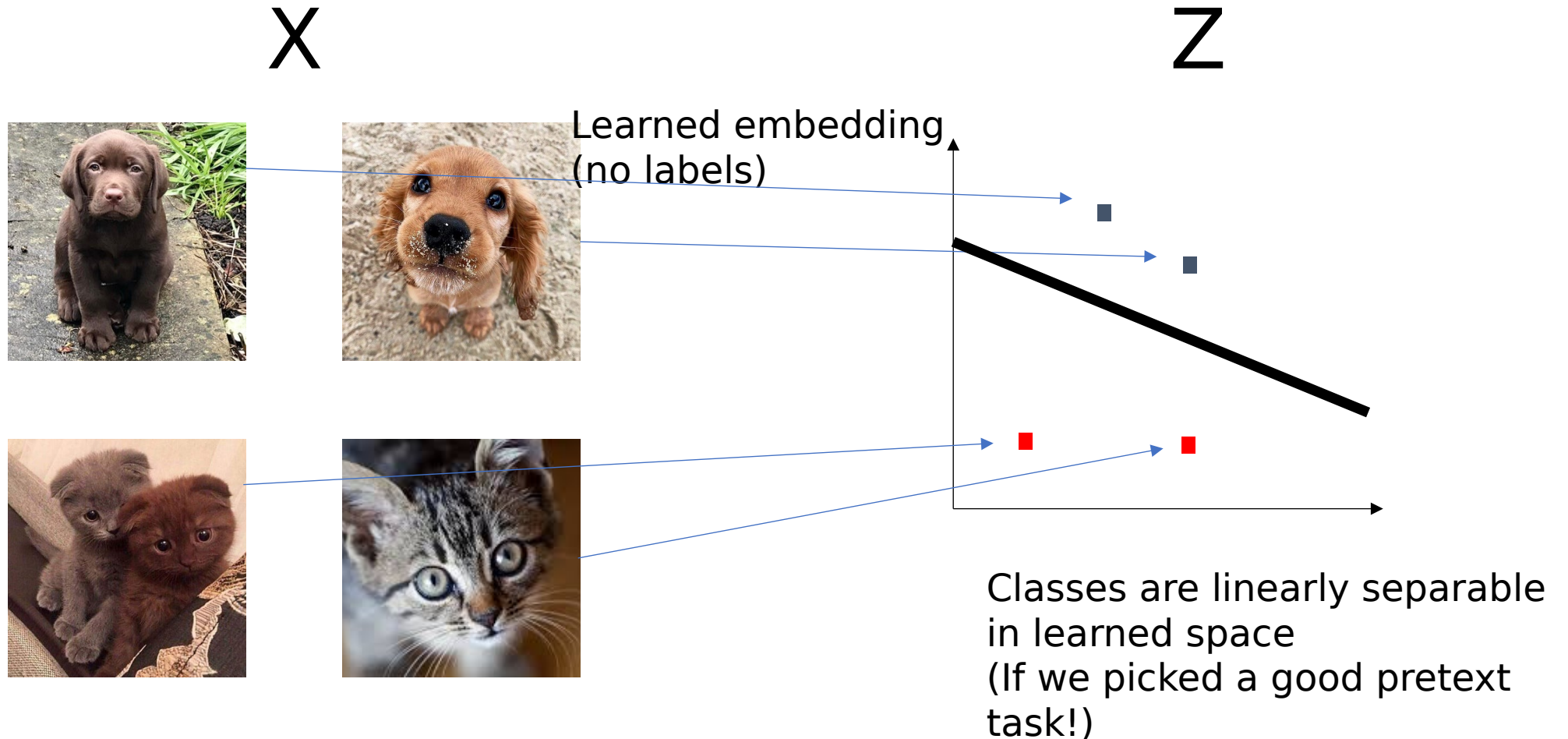
Improves adversary robustness and confidence calibration too (Wednesday?)

# Self-supervised Computer Vision

Maybe you can't pretrain on a similar large labeled dataset.  
Then you should consider pretraining on a large *self-supervised* task

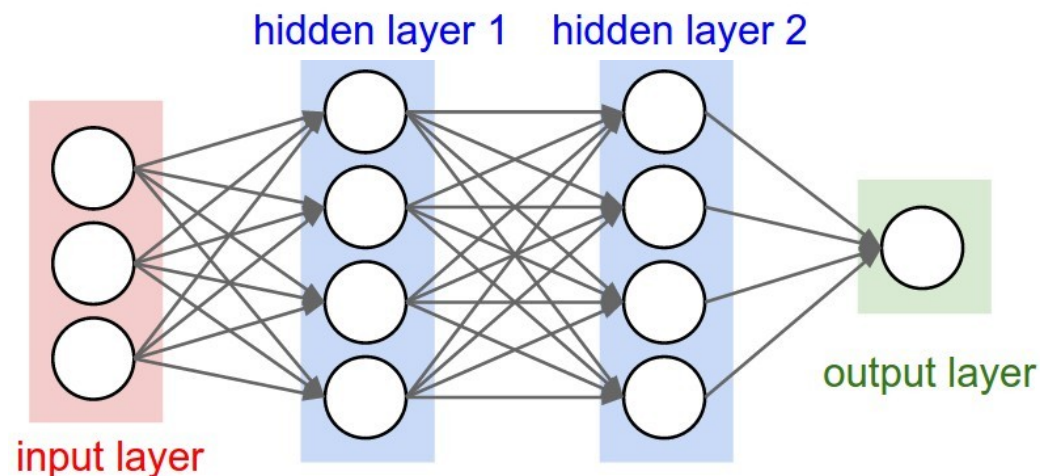
We know what to do for LLMS (next word prediction), what do we do in vision?

# Why/when does learning a self-supervised embedding help?



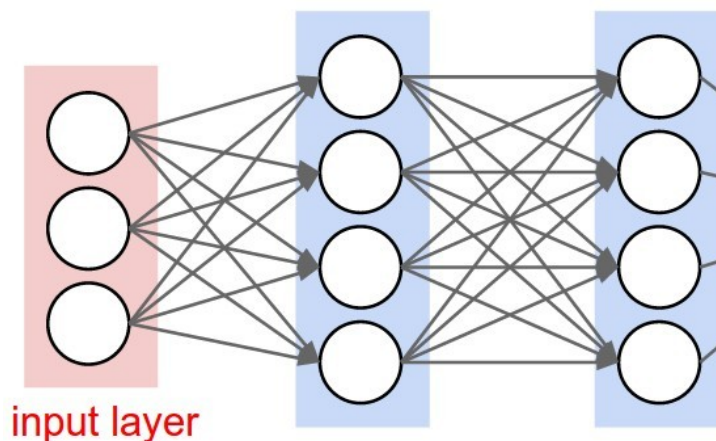


# Self-supervised pipeline evaluation



Self supervised, or "pre-text" classification task

(1) Self-supervised training



$$\hat{y} = W z + b$$

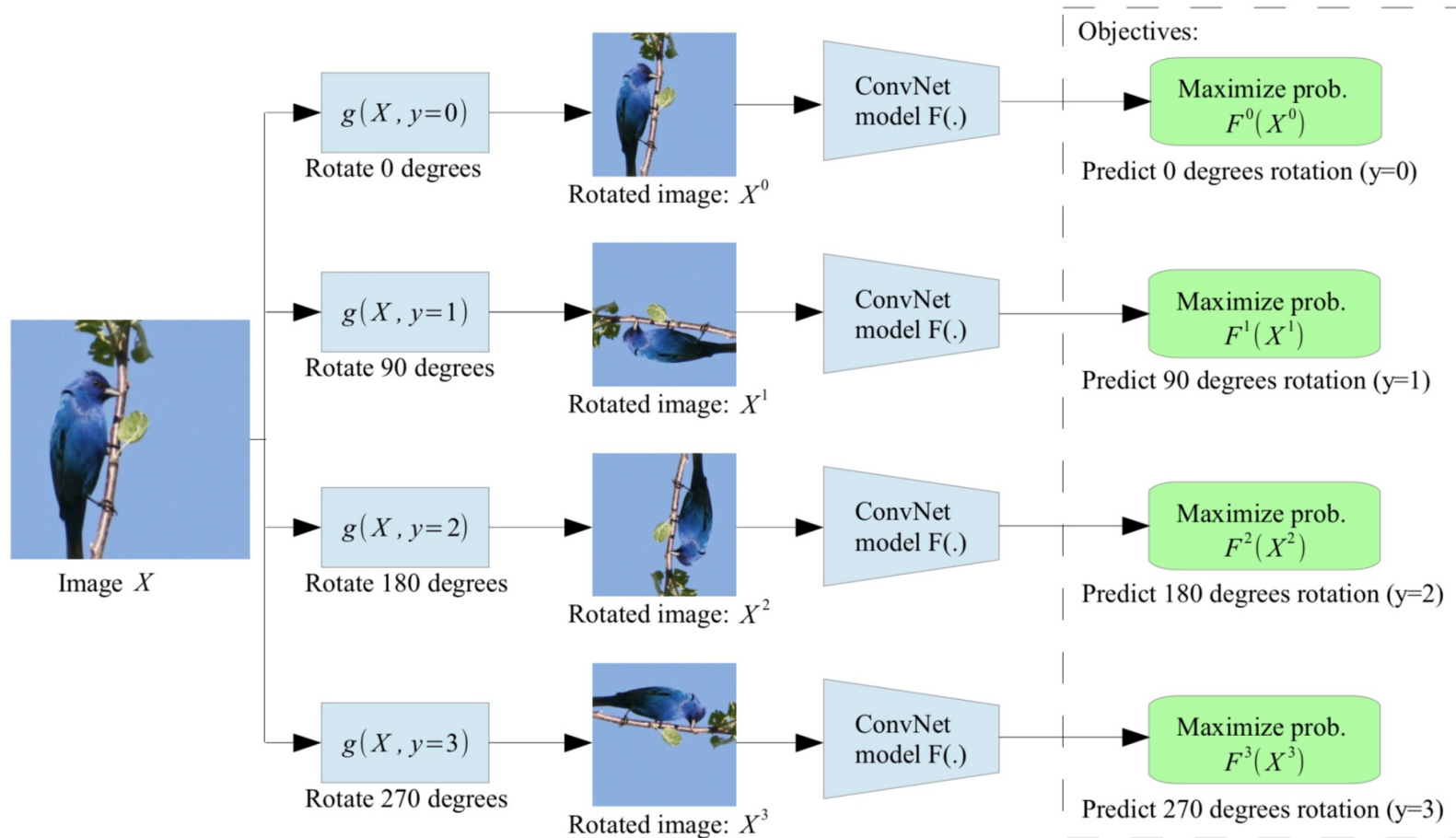
Train linear model using  $z$ , to predict target task

(2) Transfer to target task

*This is how self-supervised methods are evaluated in literature*

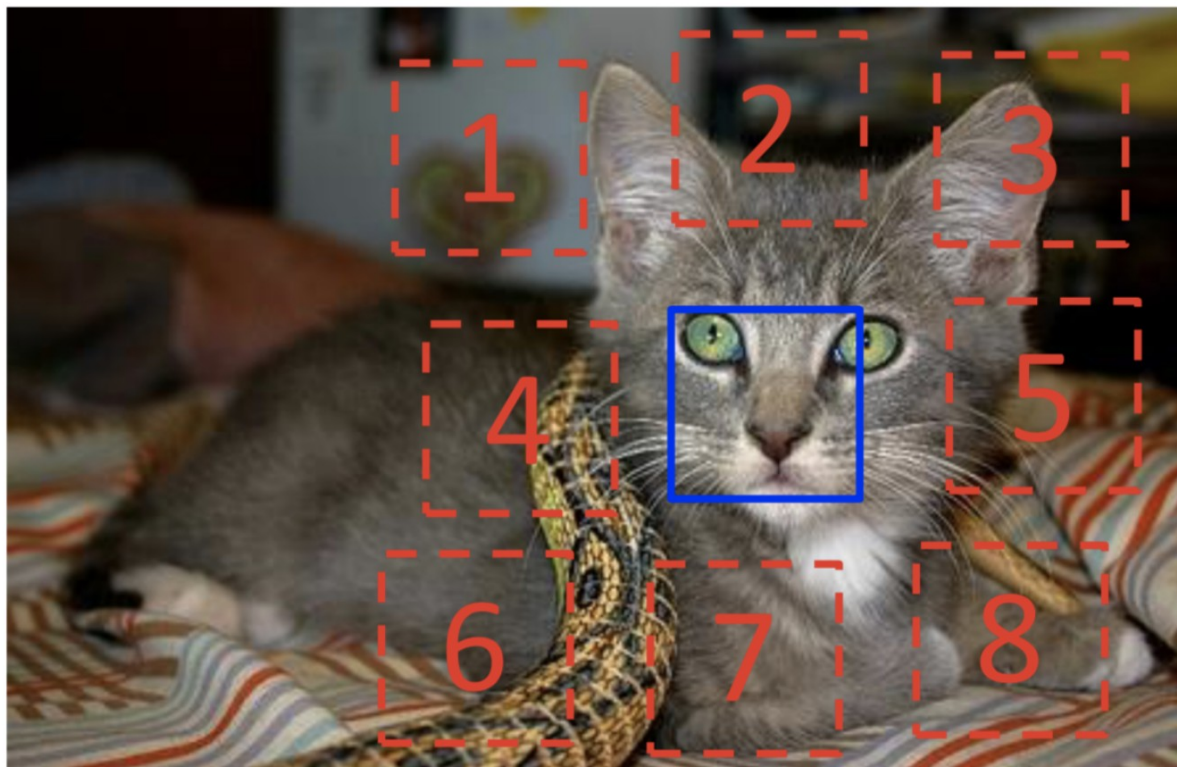
History of things people  
tried

# Predict rotations!



**Unsupervised Representation Learning by  
Predicting Image Rotations**

Spyros Gidaris, Praveer Singh, Nikos Komodakis



$$X = (\text{cat face}, \text{cat ear}); Y = 3$$

# Predict context

Example:



Question 1:



Question 2:

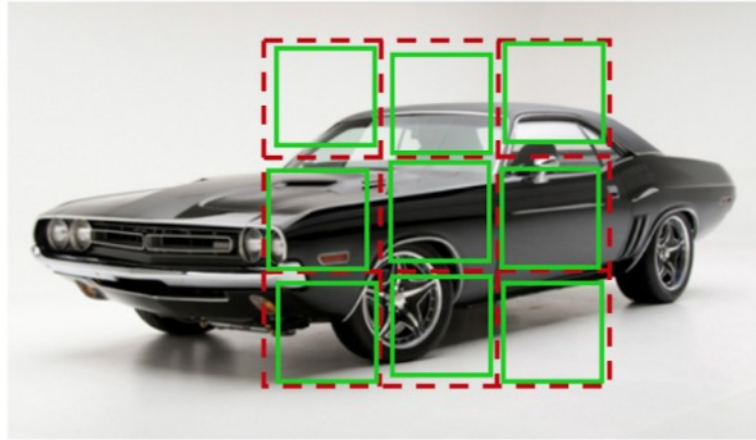


**Unsupervised Visual Representation Learning by Context Prediction**

Carl Doersch, Abhinav Gupta, Alexei A. Efros



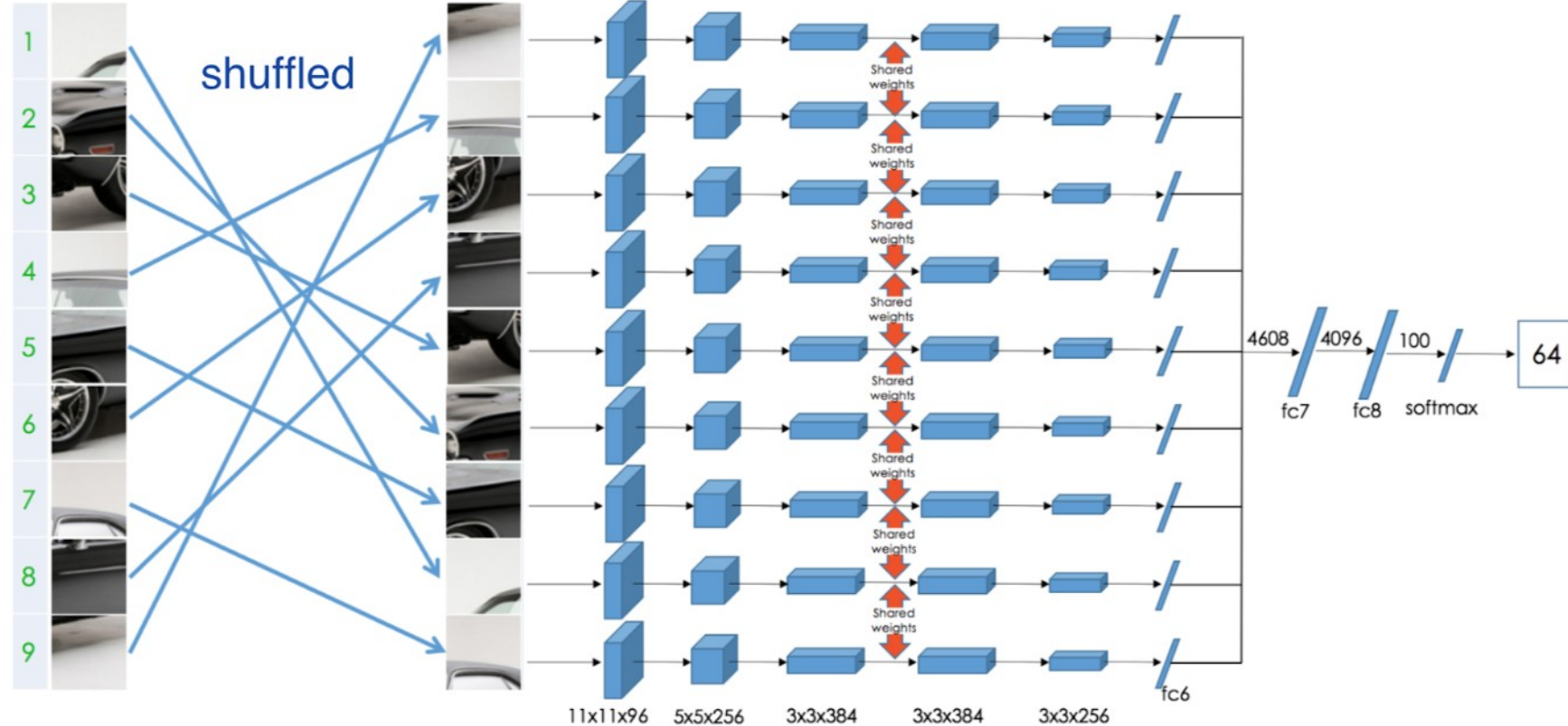
# Solve puzzles



Permutation Set

| index | permutation       |
|-------|-------------------|
| 64    | 9,4,6,8,3,2,5,1,7 |
|       |                   |
|       |                   |

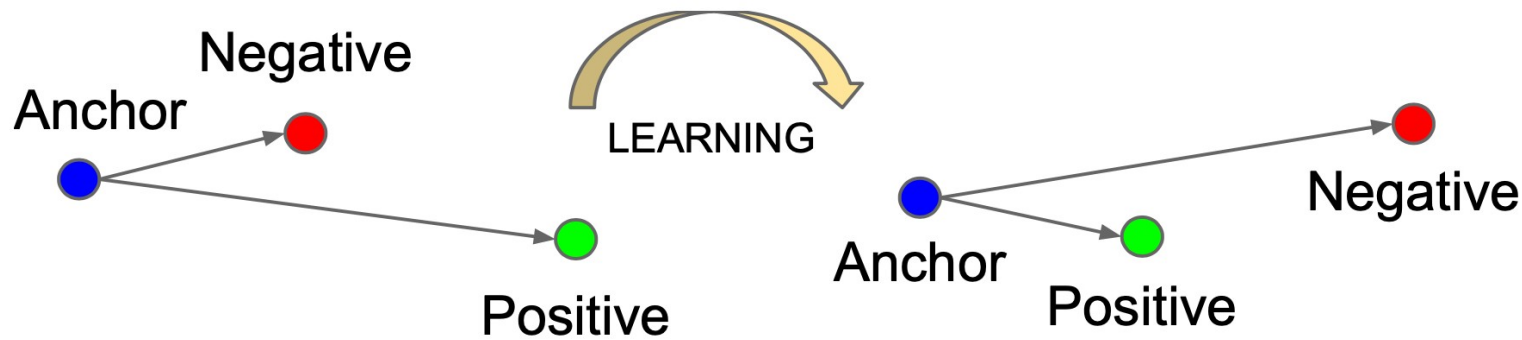
Reorder patches according to the selected permutation



## Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

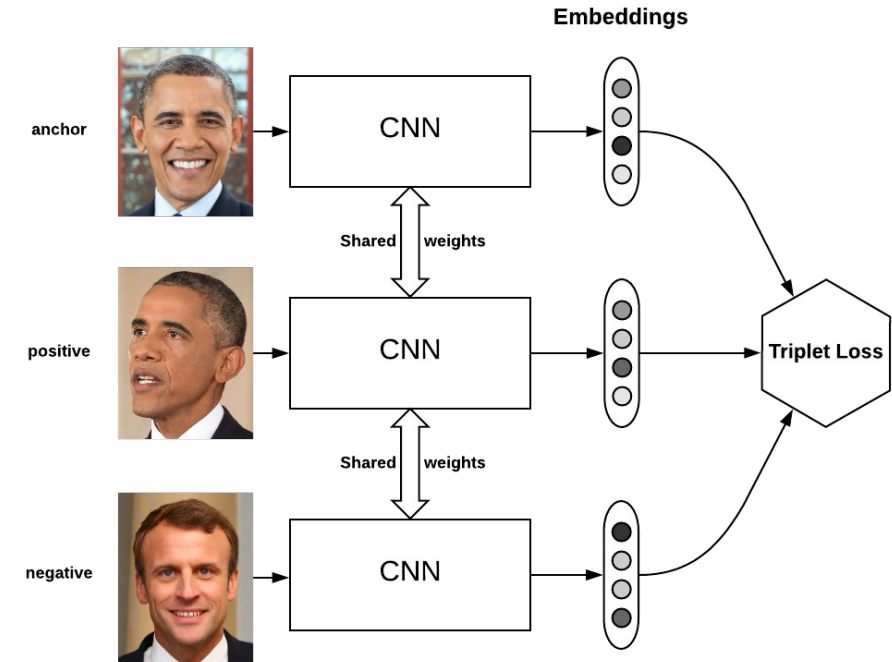
Mehdi Noroozi, [Paolo Favaro](#)

# Contrastive learning – triplet loss



$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max(0, \|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2 - \|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2 + \epsilon)$$

Make these close Make these far apart



To do it in a more self-supervised way, the positive could be a modification of the anchor image, and the negative could be

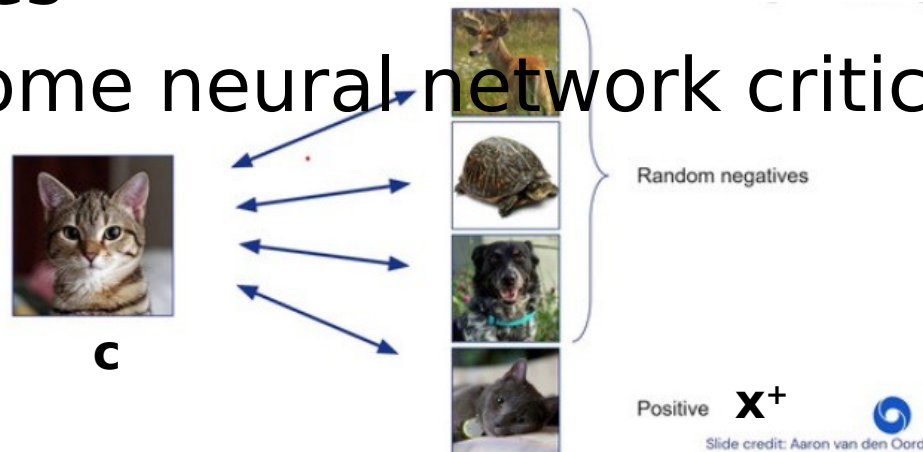
# More popular/modern form of contrastive loss: InfoNCE / CPC

“Info Noise Contrastive Estimation”, “Contrastive Predictive Coding”

Original paper: <https://arxiv.org/abs/1807.03748>

Predict a match in a batch with one positive and many negatives

Using some neural network critic function,  $f$ , a classifier



$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[ \log \frac{f(\mathbf{x}^+; \mathbf{c})}{\sum_{\mathbf{x}' \in X} f(\mathbf{x}', \mathbf{c})} \right]$$

(Often something like cosine similarity between  
embeddings)

# Contrastive loss math

We want to define a probability distribution – predict which of the images on the right is correctly matched with left.

First, use a similarity measure. Dot product is common:

$$\text{Sim}(a, i) = z_a \cdot z_i$$

How do we turn it into a distribution?

$$p(y = j) = \text{softmax} \text{sim}(z_a, z_j) = \frac{e^{z_a \cdot z_j}}{\sum_i e^{z_a \cdot z_i}}$$

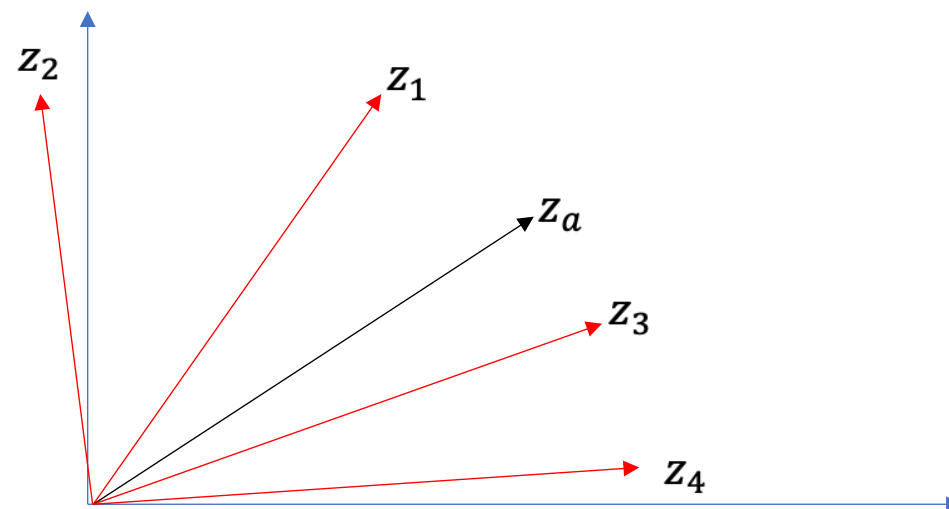
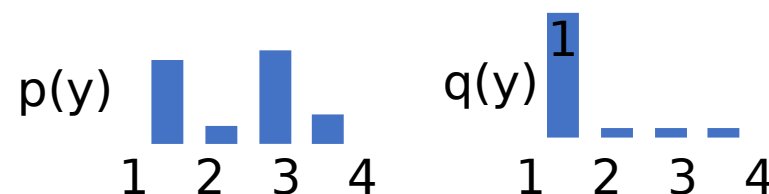
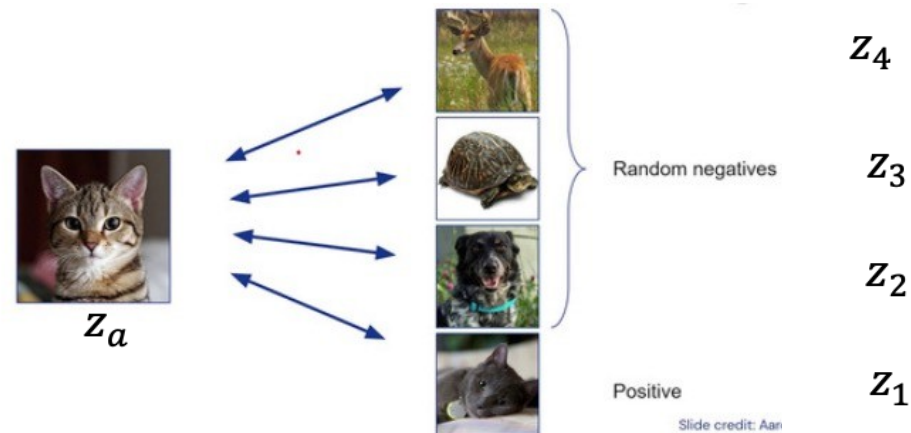
Now, we have to compare this to the *true* answer.

We have set it in this case,  $y=1$ , or  $q(y=1) = 1$ ,

$q(y \neq 1) = 0$ .

Cross entropy:

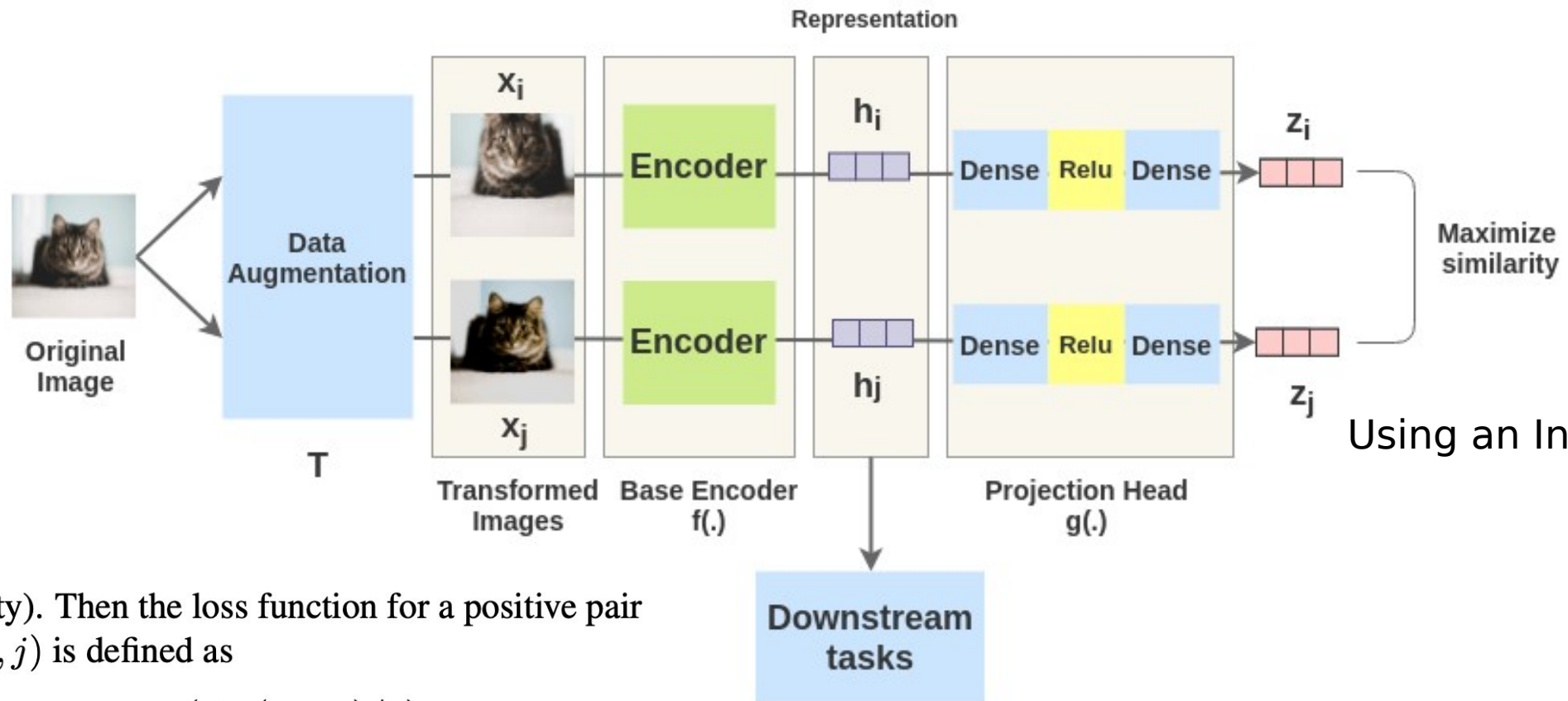
$$H(p, q) = -\sum_y q(y) \log p(y) = -\log \frac{e^{z_a \cdot z_1}}{\sum_i e^{z_a \cdot z_i}}$$





# SimCLR – Self-supervised contrastive similarity learning

SimCLR Framework



Using an InfoNCE style loss

cosine similarity). Then the loss function for a positive pair of examples  $(i, j)$  is defined as

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

# SimCLR augmentations



(a) Original



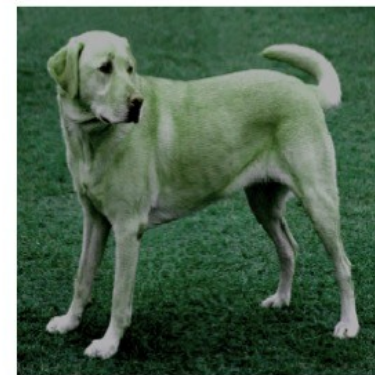
(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate  $\{90^\circ, 180^\circ, 270^\circ\}$



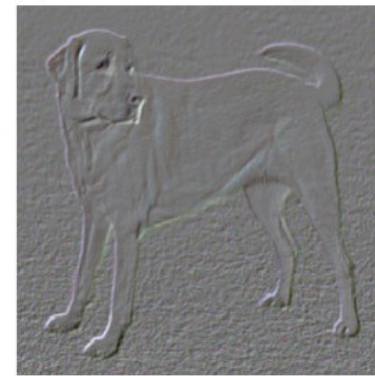
(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering

# SimCLRv2

## Big Self-Supervised Models are Strong Semi-Supervised Learners

Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, Geoffrey Hinton  
Google Research, Brain Team

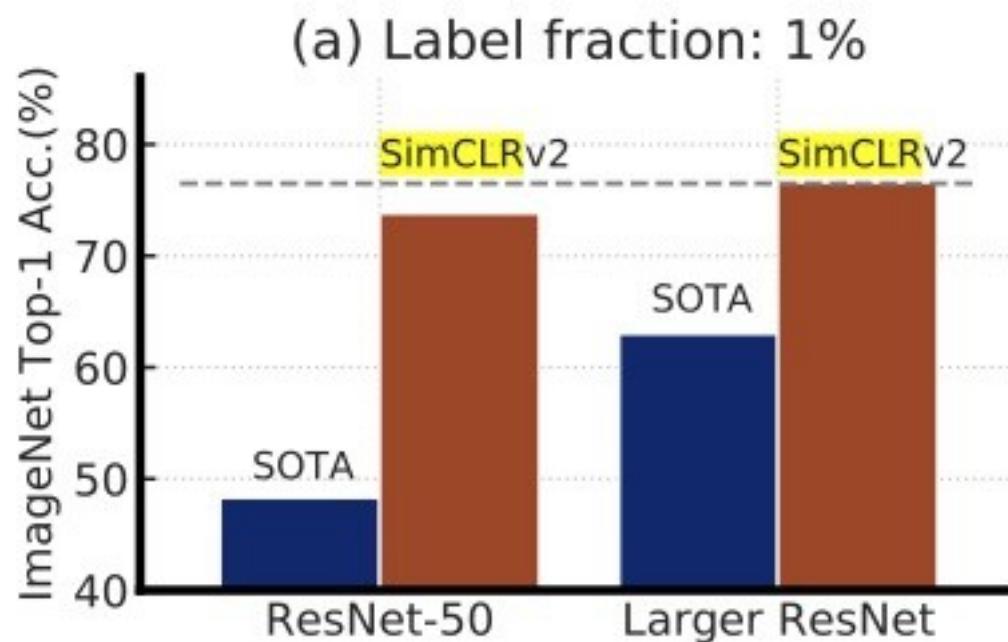


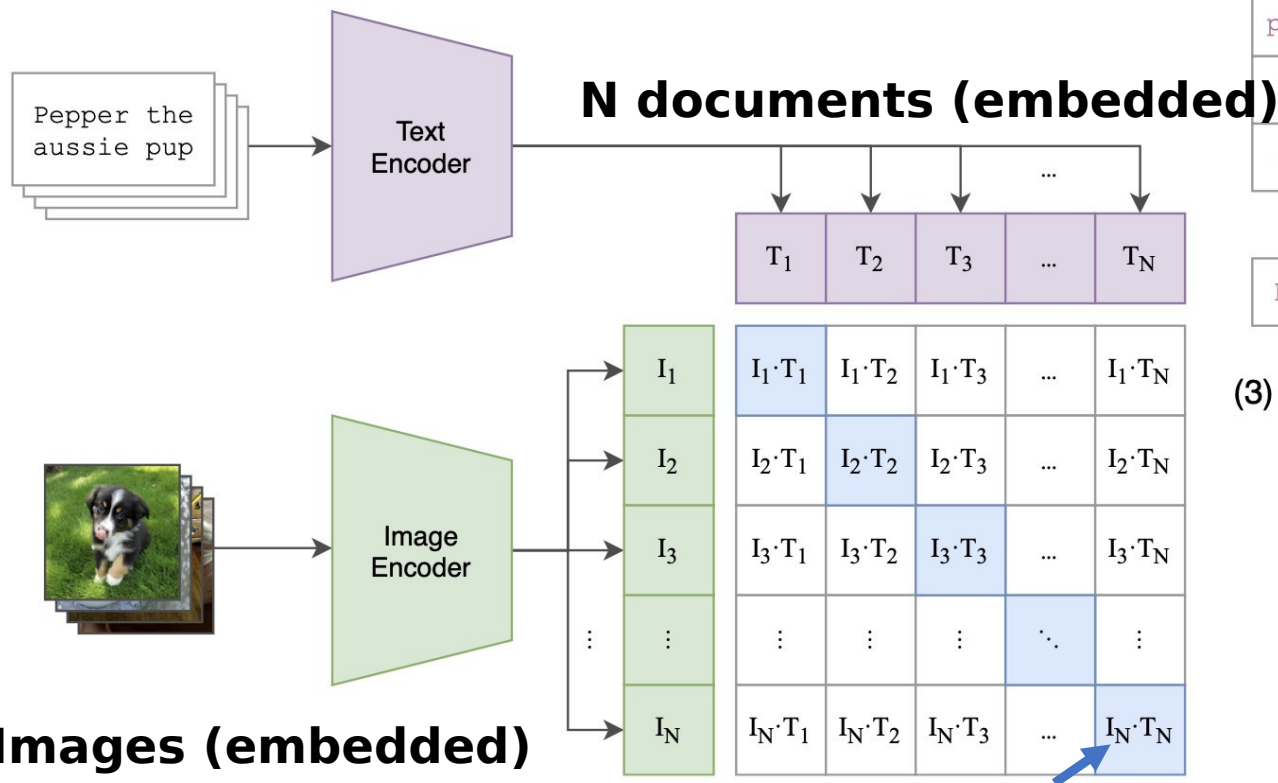
Figure 2: Top-1 accuracy of previous state-of-the-art (SOTA) methods [1, 2] and our method (SimCLRv2) on ImageNet using only 1% or 10% of the labels. Dashed line denotes fully supervised ResNet-50 trained with 100% of labels. Full comparisons in Table 3.

Quick aside

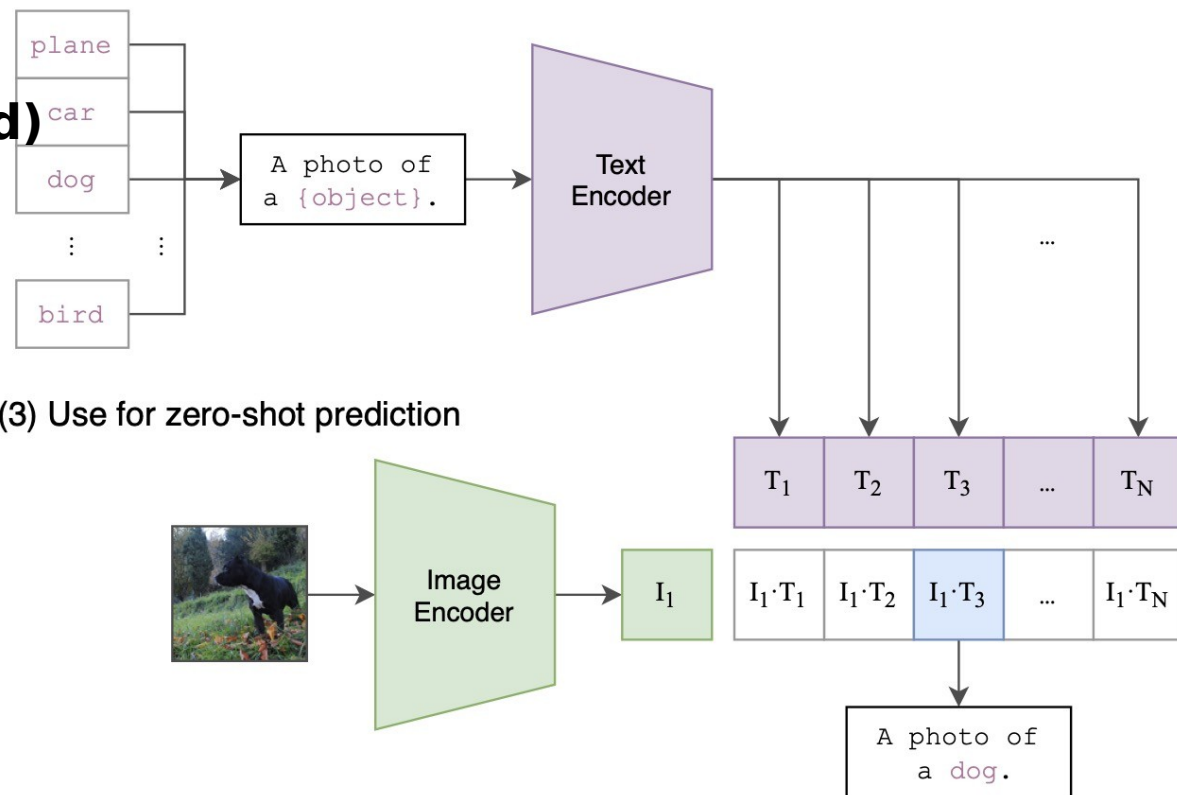
**CLIP: Contrastive  
Language-Image  
Pretraining**



(1) Contrastive pre-training



(2) Create dataset classifier from label text



*Figure 1.* Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset’s classes.



## Learning Transferable Visual Models From Natural Language S

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t            - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

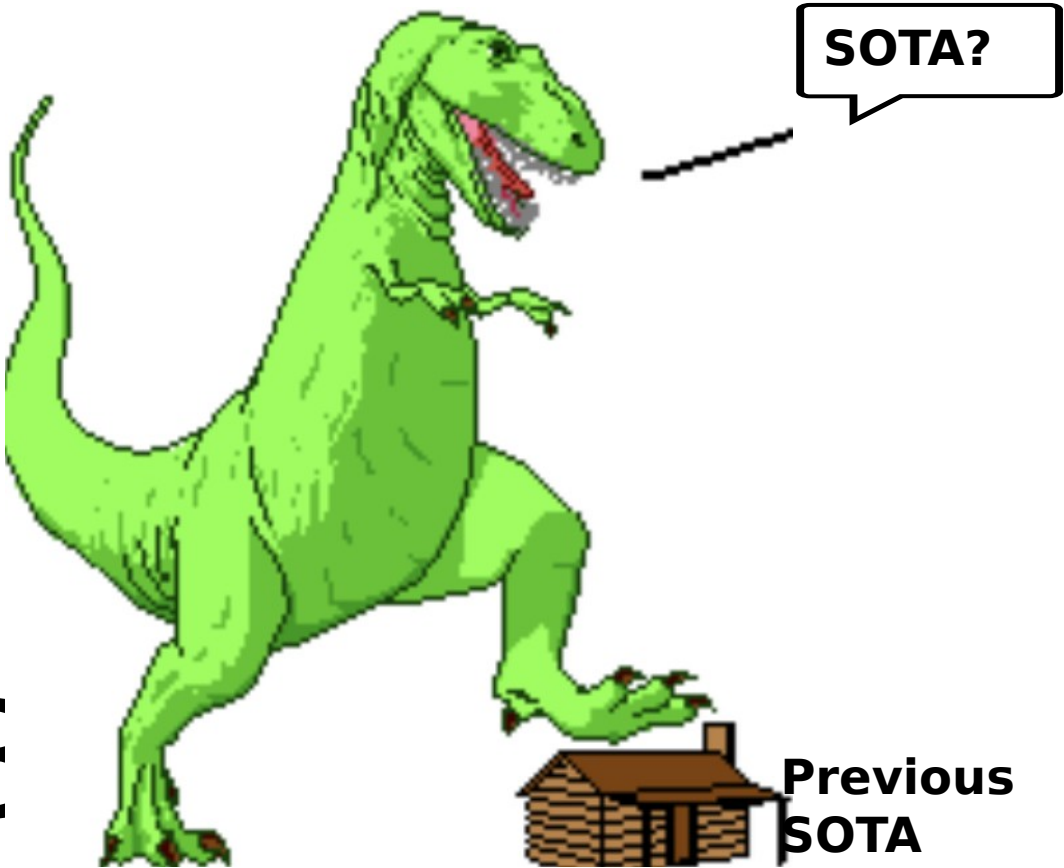
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

*Figure 3.* Numpy-like pseudocode for the core of an implementation of CLIP.

DINO (v1-v $\infty$ )



Previous  
SOTA

# DINO v1: Self-Distillation w **NO** Labels

Vision transformer backbone – attention learns about objects without labels!

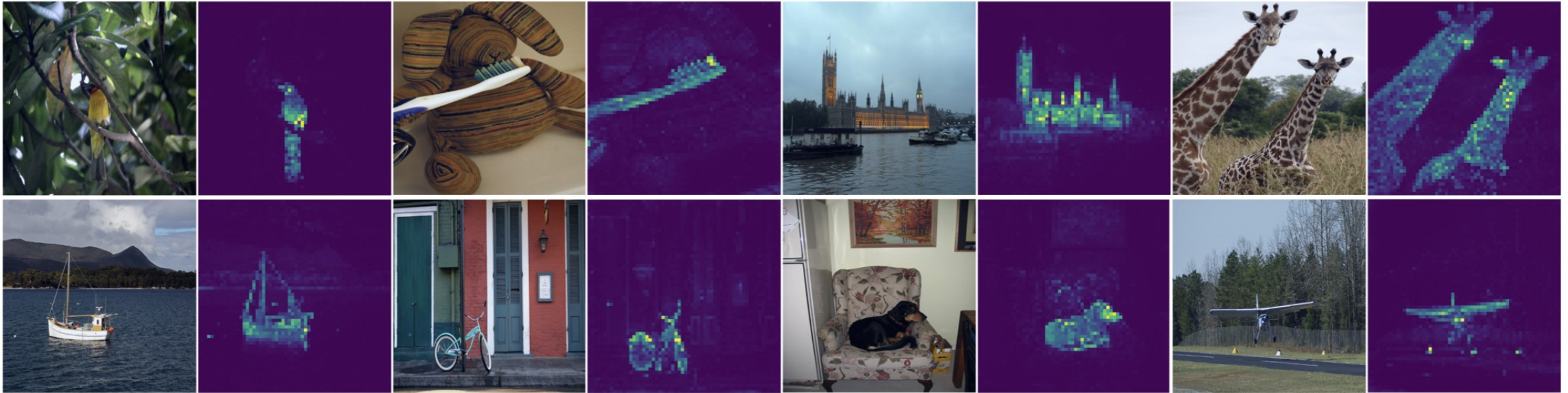
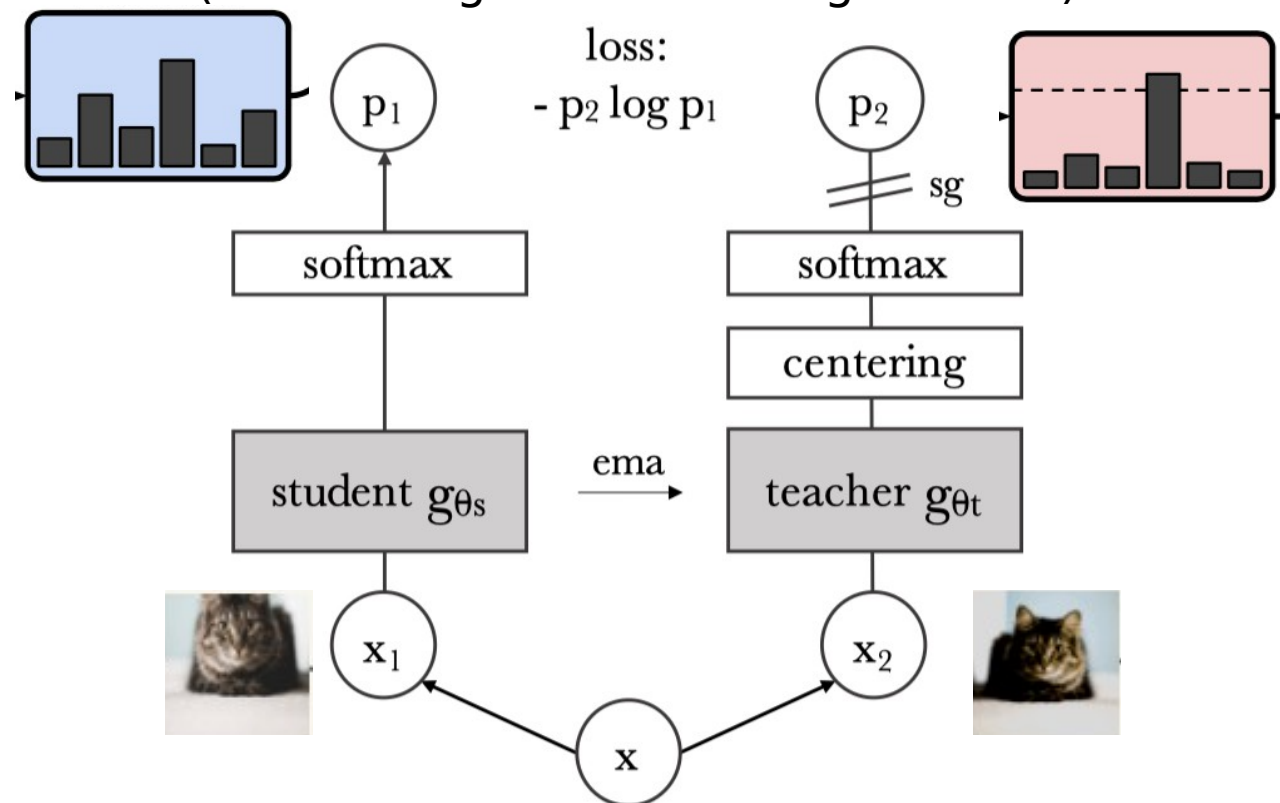


Figure 1: **Self-attention from a Vision Transformer with  $8 \times 8$  patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

## Predict same “label” distribution

(even though we have no g.t. labels)



**Augmented views** just like before



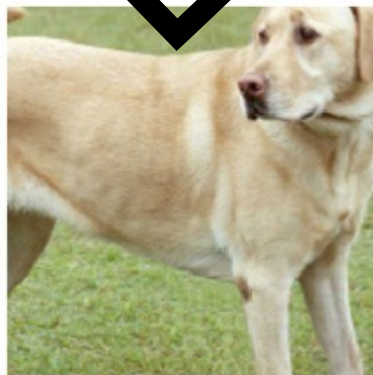
# Mostly same augmentation as SimCLR



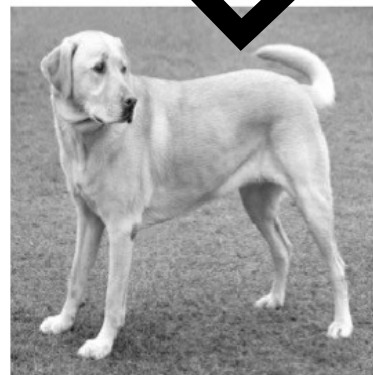
(a) Original



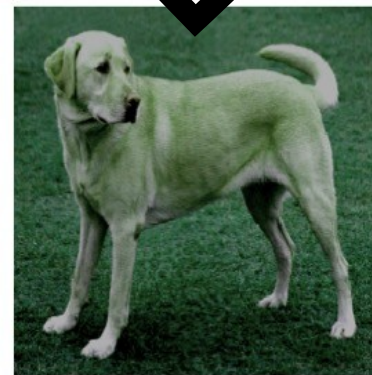
(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate  $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



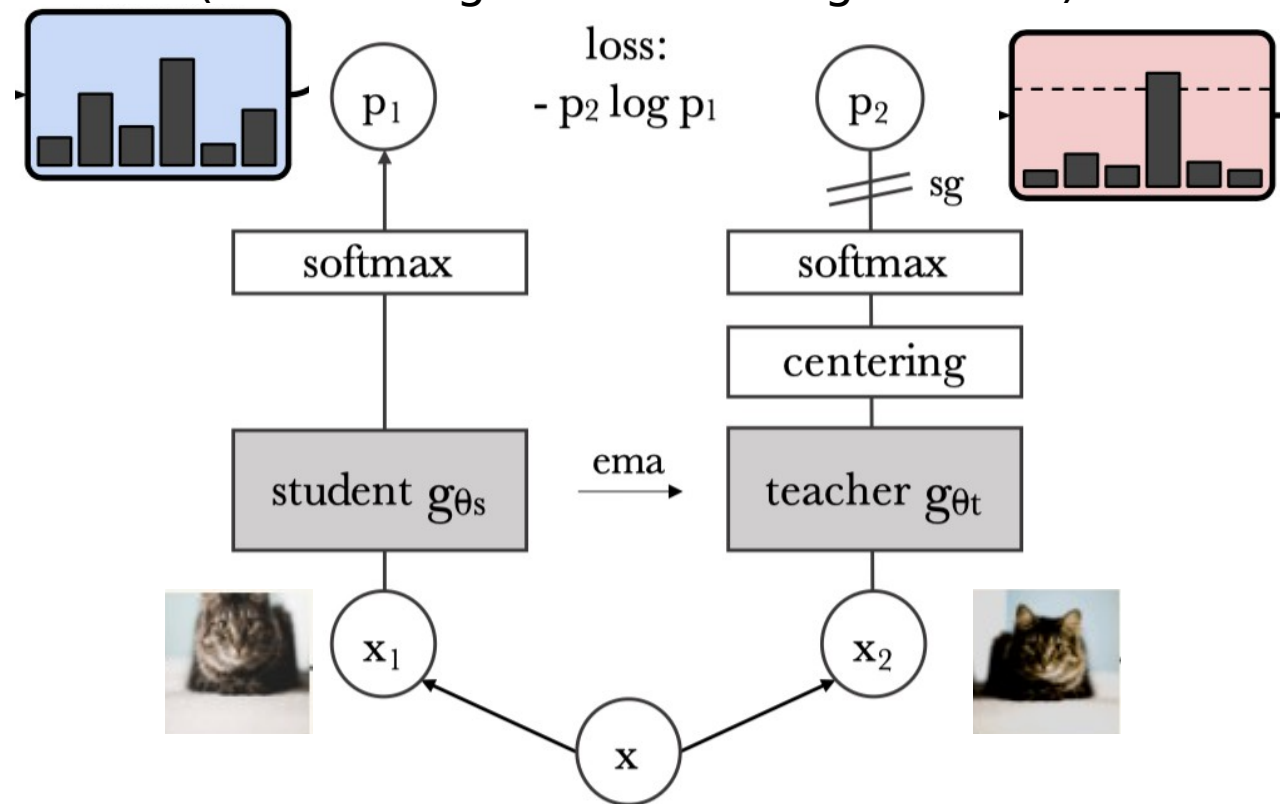
(j) Sobel filtering

+ "solarization"



“self-  
distillation”?  
Teacher =  
Exponential  
Moving  
Average of  
Student?

**Predict same “label” distribution**  
(even though we have no g.t. labels)



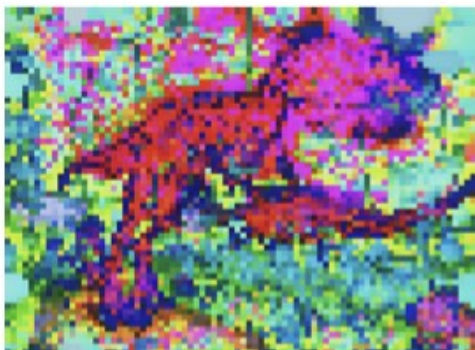
**Augmented views** just like before

# v2-3: mostly scaling improvements

**Improve the per-pixel or "dense" features**



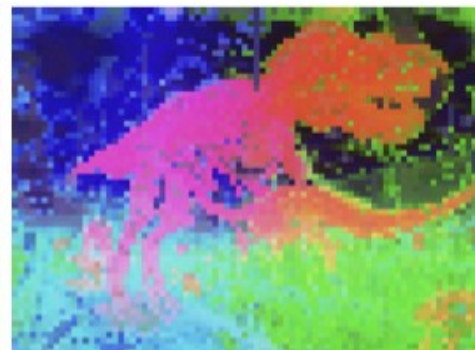
Input



SigLIP 2



PE Spatial



DINOv2 w/reg



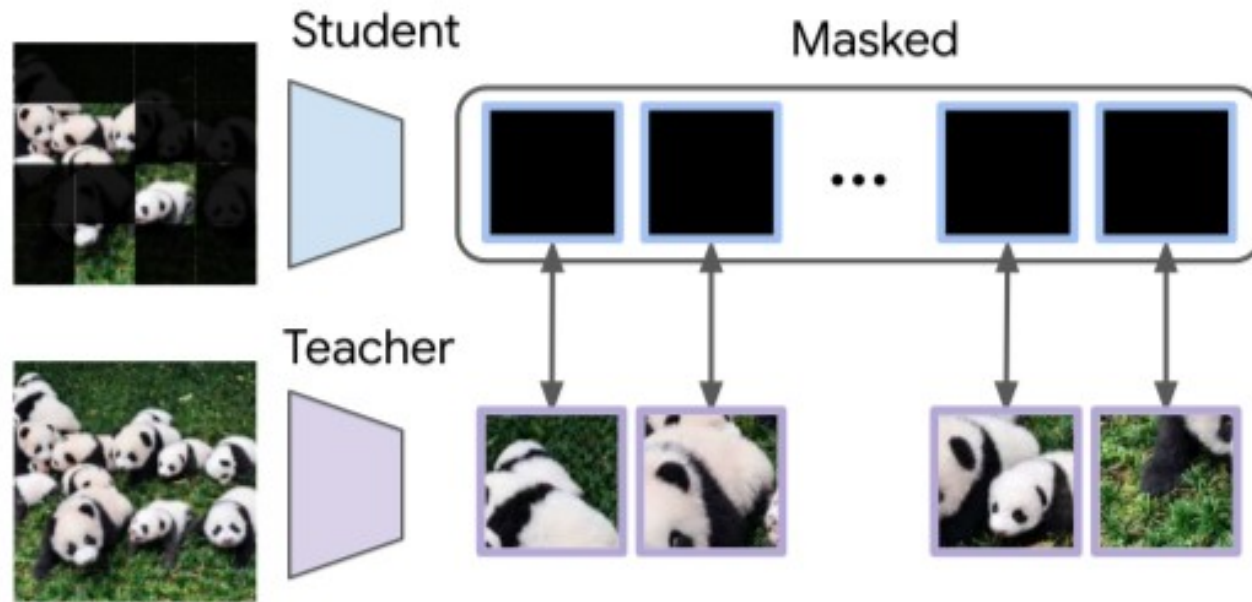
DINOv3



# Adding patch-level losses

- Student-teacher loss on Patch Features – “Gram Anchoring”
- And iBOT

iBOT (prev.): Self-Distillation on Masked Patches





# Problems with vision and uncertainty

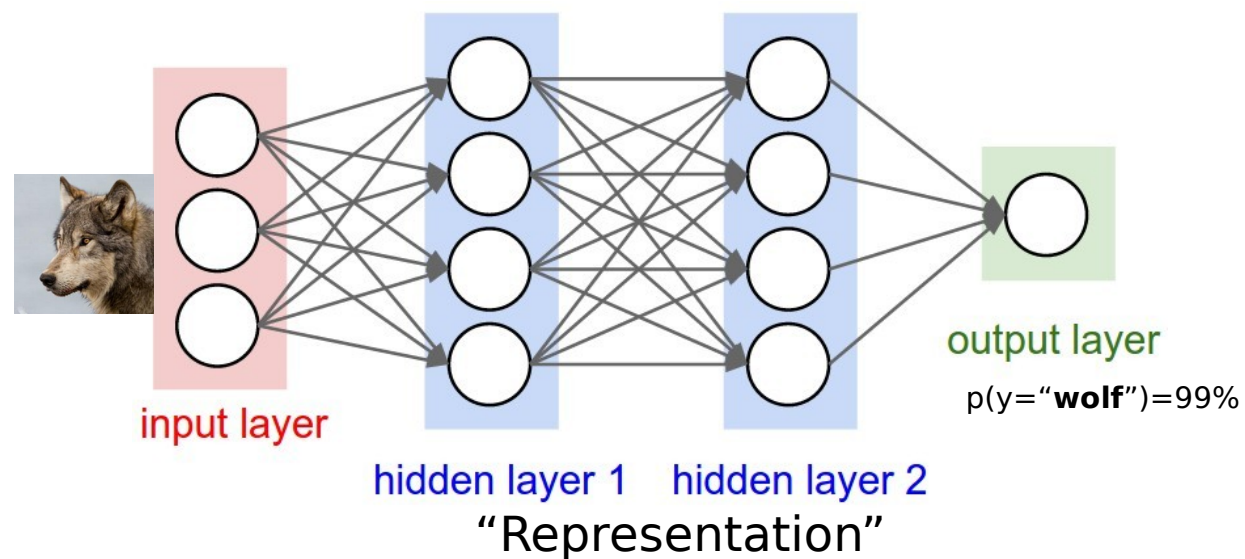
Lead in to discussing *uncertainty*



$p(y=\text{"wolf"})=99\%$



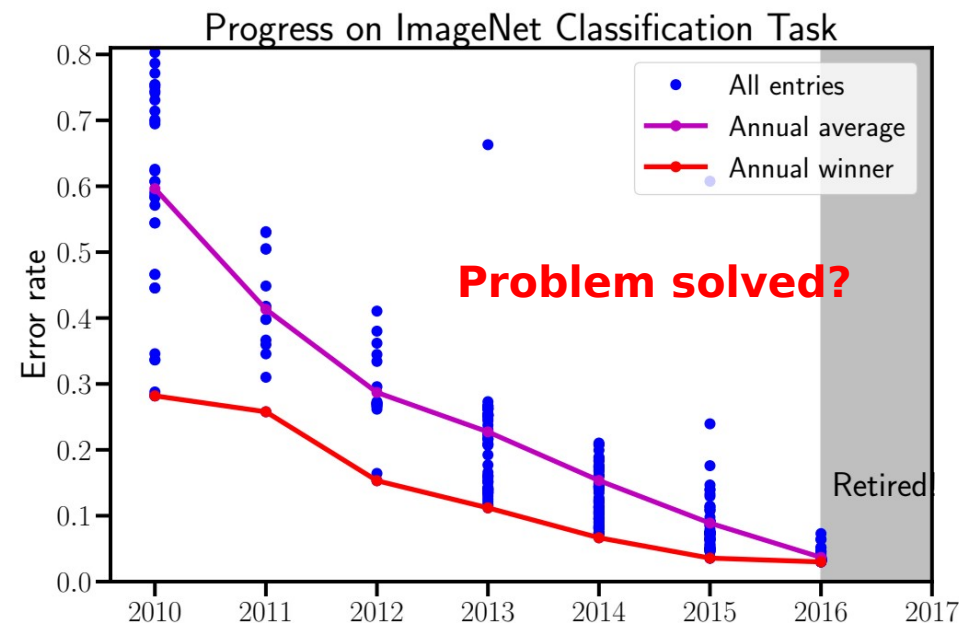
$p(y=\text{"lion"})=99\%$



***Image classification benchmarks were easy and artificial***

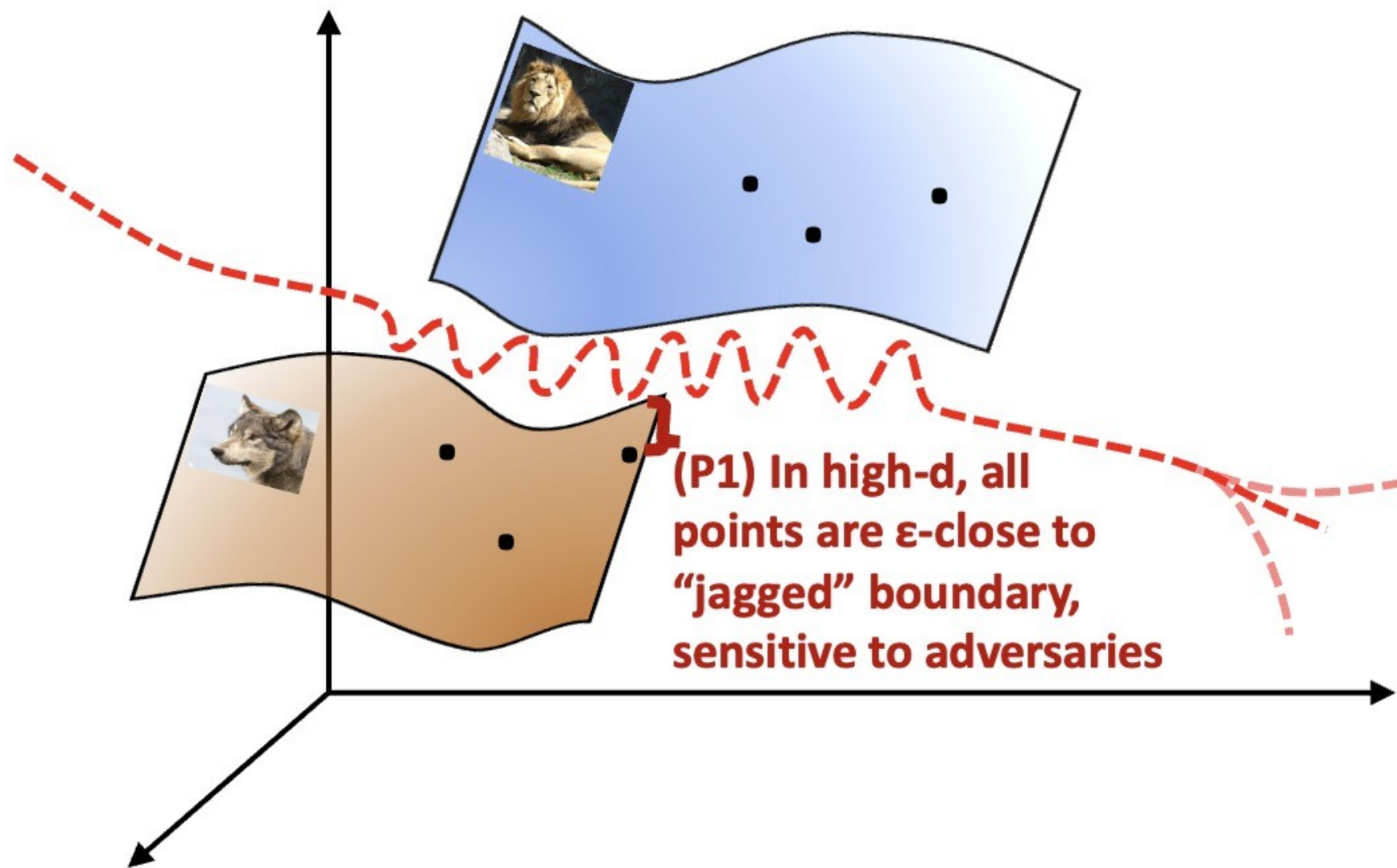
***Information used by neural nets and humans is not the same!***

***Hard problems remain.***



# Sensitivity to adversarial perturbations





x-space (image pixels)



# Real world attacks



<https://arxiv.org/pdf/1707.08945.pdf>

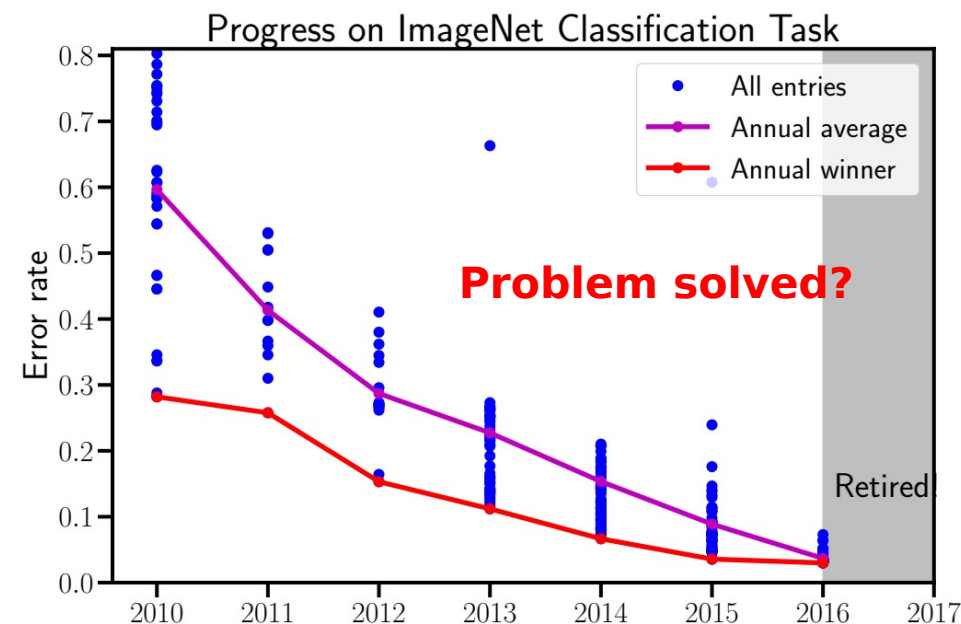
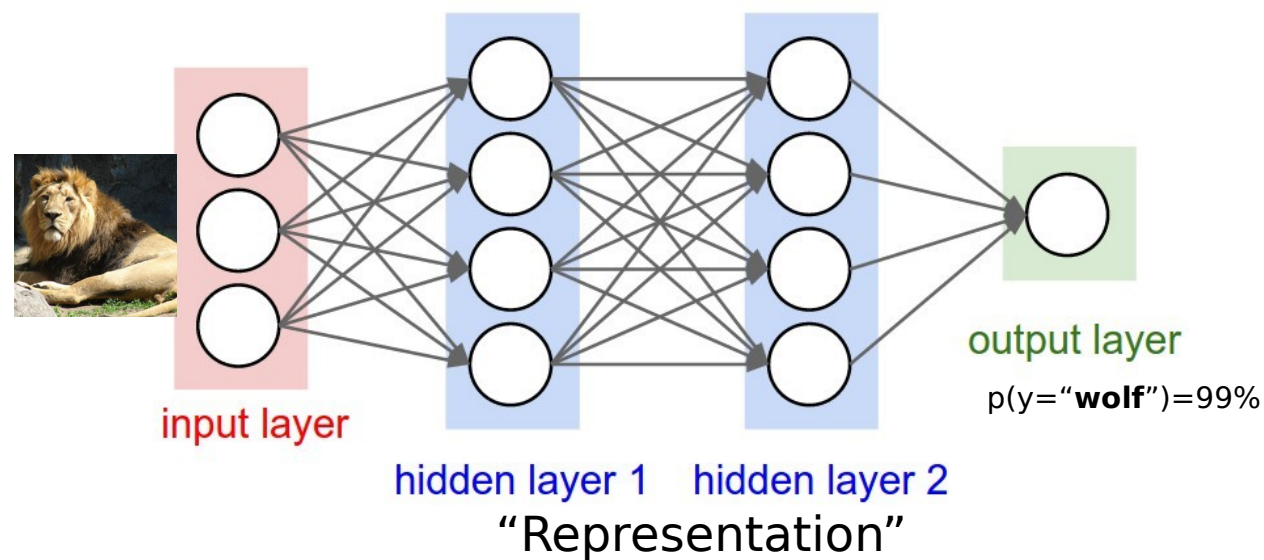
# Another issue – far Out Of Distribution (OOD) inputs

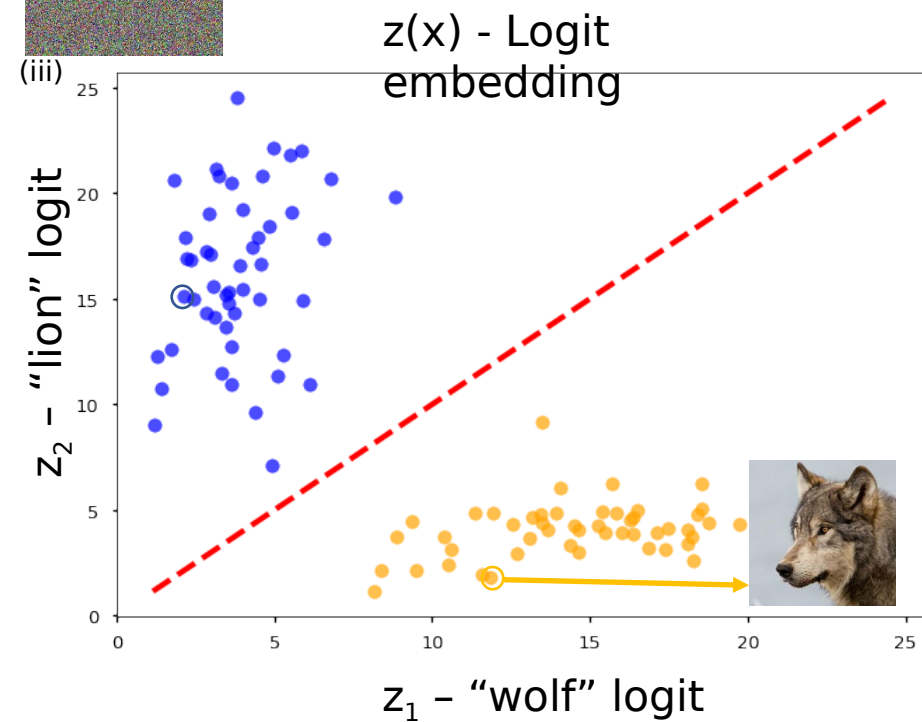
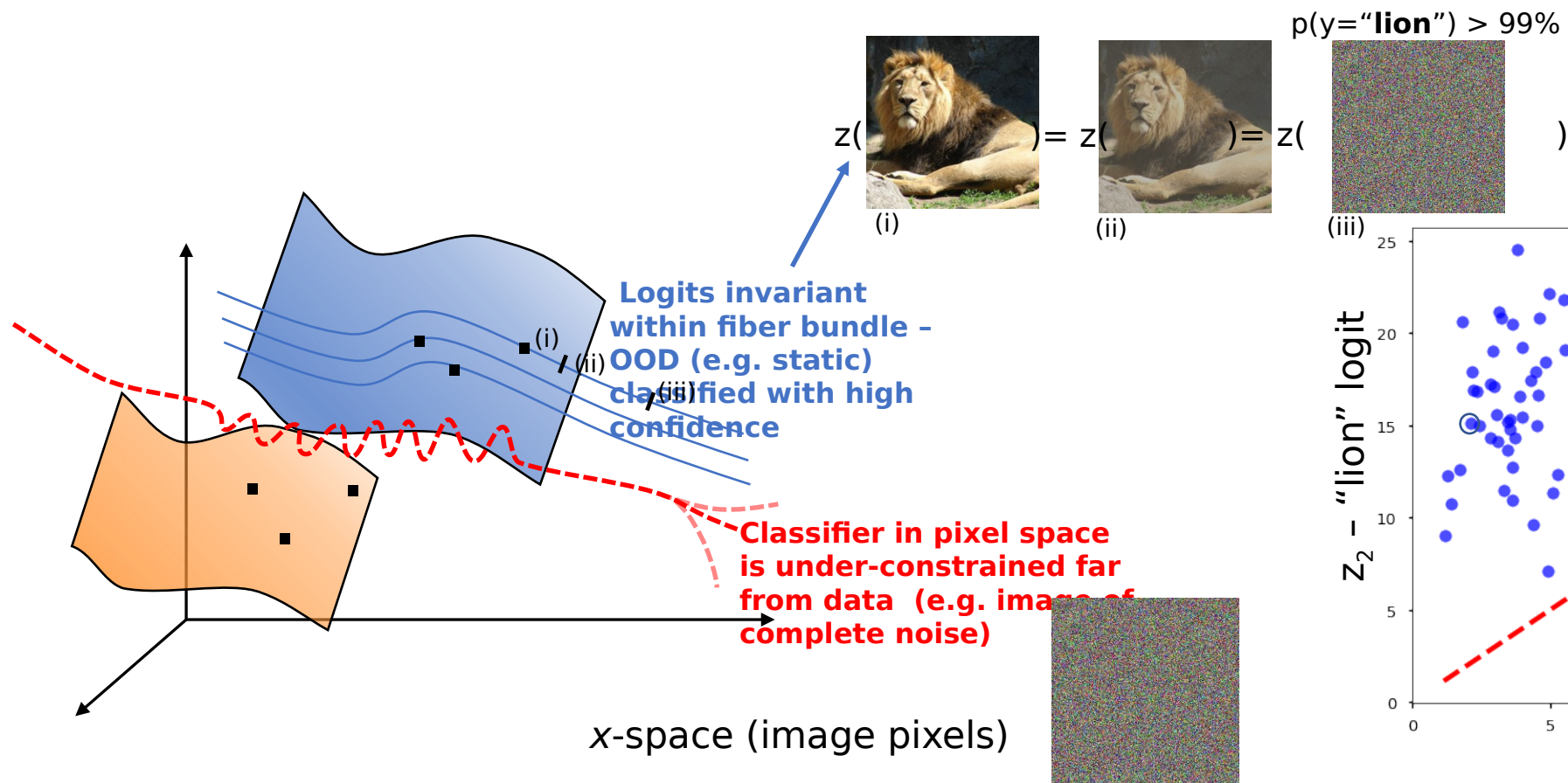


$p(y=\text{"lion"})=99\%$



$p(y=\text{"lion"})=99\%$

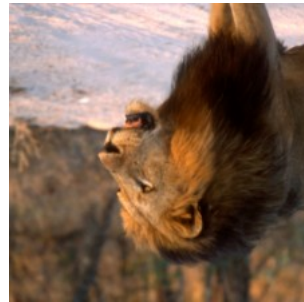




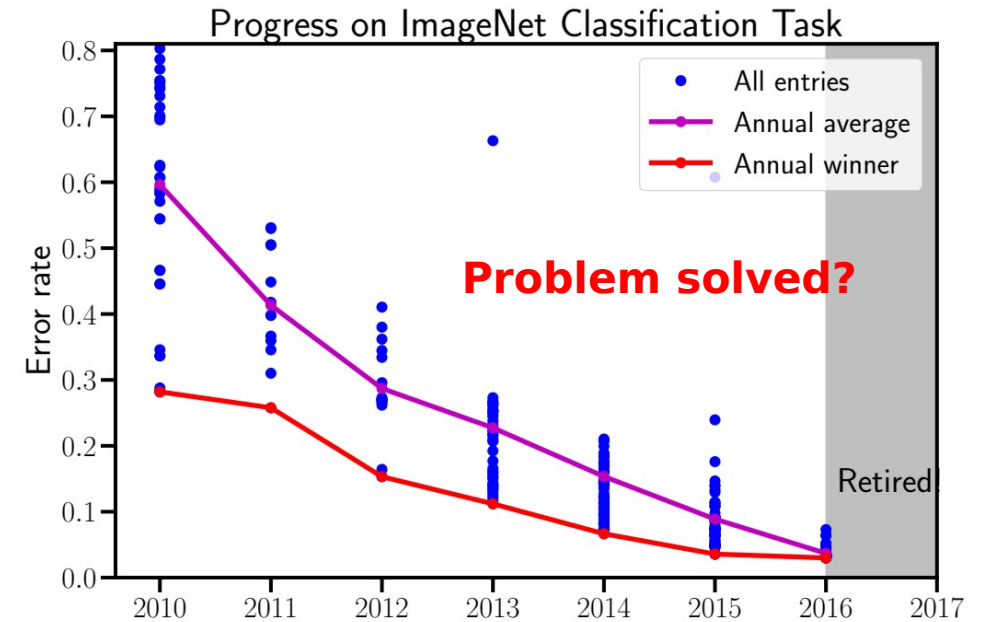
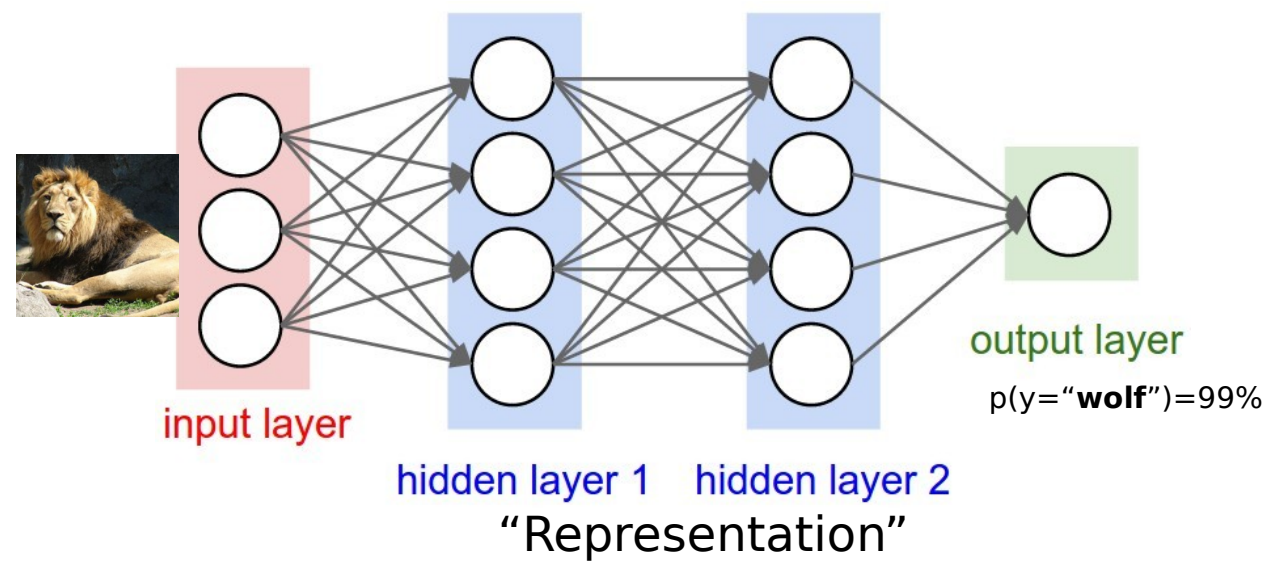
# Predictions not invariant to “nuisance” variations



$\hat{y}$ ="lion"

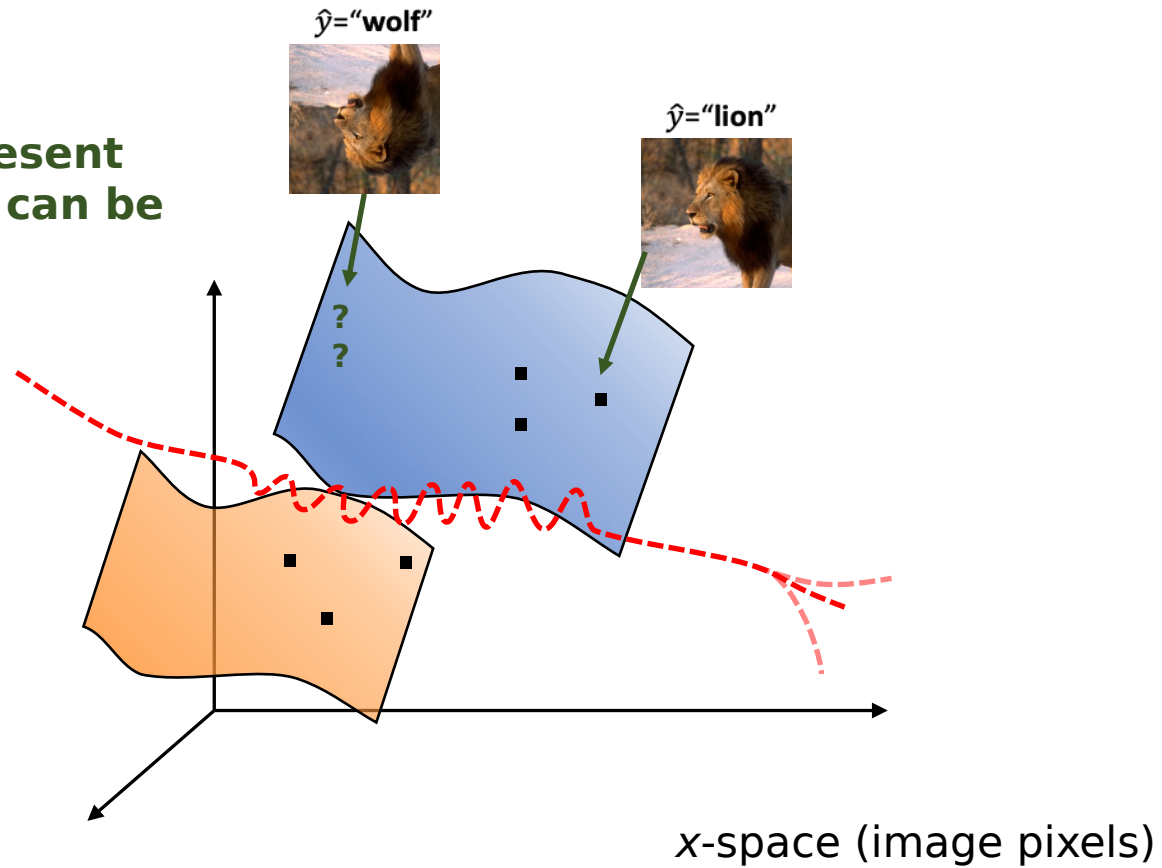


$\hat{y}$ ="wolf"



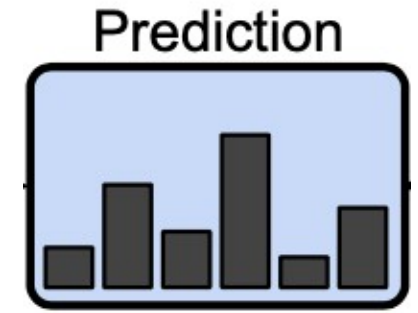


**Parts of image manifold not present in training data can be misclassified**

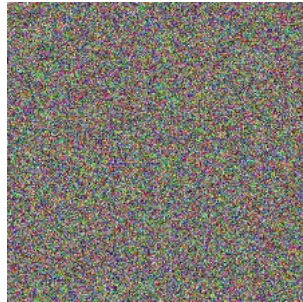


We attempt to mitigate some of these issues with semi- and self-supervised learning

# What do the probabilities that our neural nets give us really *mean*?



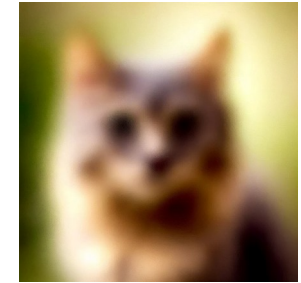
$p(y=\text{"lion"})=99\%$



$p(y=\text{"lion"})=99\%$



$p(y=\text{"lion"})=99\%$



$p(y=\text{"cat"})=80\%$

We often say that the probability a neural net assigns to the top choice is its **“confidence”**. If a neural net is 80% confident of a prediction, will that prediction be correct 80% of the time?

Confidence calibration game: give a range that you think the answer lies in with probability 50%

Martin Luther King's age at death:

Length of the Nile River:

Number of countries that are members of OPEC:

Number of books in the Old Testament:

# My guesses

## Correct answers

Martin Luther King's age at death:

35-55

**39**

Length of the Nile River:

700-1500 miles

**6737 km or 4187 miles**

Number of countries that are members of OPEC:

9-23

**12 (as of 2026)**

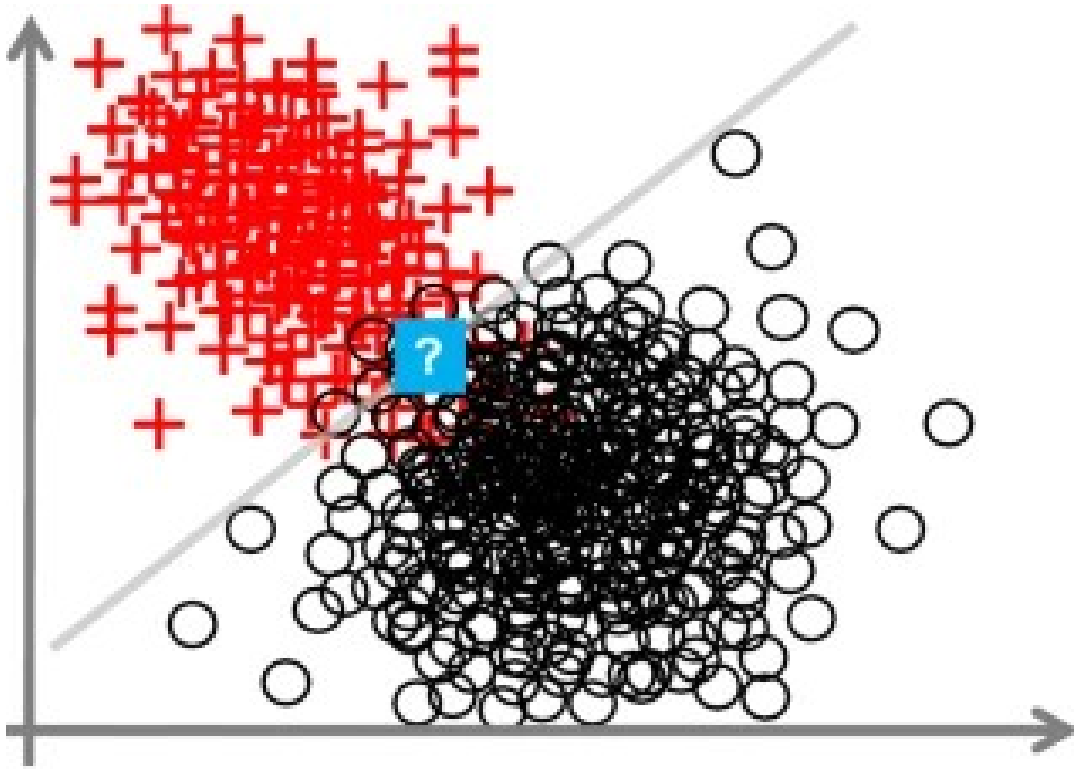
Number of books in the Old Testament:

30-55

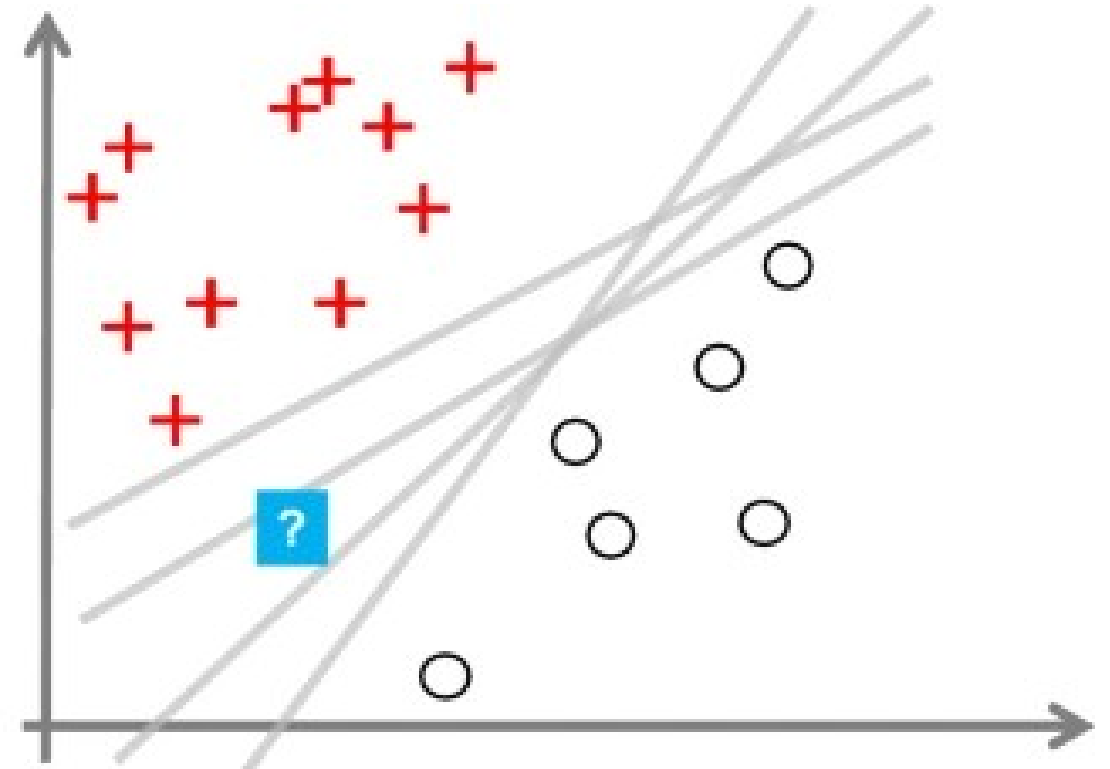
**39**



# Two types of uncertainty



**aleatoric / fundamental / irreducible / data uncertainty**



**Epistemic / Model uncertainty**  
- due to our lack of knowledge

Do we care? Lots of applications. Right side will cause problems.